Evaluation of the dCGF Model

Qian et al. (2025) bioRxiv 2025.01.04.631316

Reproduction using KEGG pathways (144) instead of BioCyc (500)
Clark et al. 2021 dataset: 25 species, equal initial abundances
dCGF-IS (Individual Species) model, 8-fold CV, 5 seeds

Key Findings

1. Model reproduces paper results with fewer pathways
   (our 144 KEGG pathways vs paper's 500 BioCyc pathways)

2. Shuffled-target permutation test confirms real biological
   signal (r ~ 0 for all metabolites)

3. Species-only model (no pathway data) achieves r = 0.56 - 0.88
   -> species composition is the dominant predictor

4. Random-abundance control achieves r = 0.52-0.82:
   binary pathway vectors encode species identity, limiting
   the information removed by abundance randomization

5. Sensitivity analysis (Figure 4) is influenced by pathway
   prevalence: rare pathways act as species markers

6. Predictions are driven largely by community composition;
   pathways provide additional but modest improvement

Paper Figure 2: dCGF Model Architecture
dCGF-ES (Enzyme Soup) sums across species; dCGF-IS (Individual Species) preserves per-species vectors

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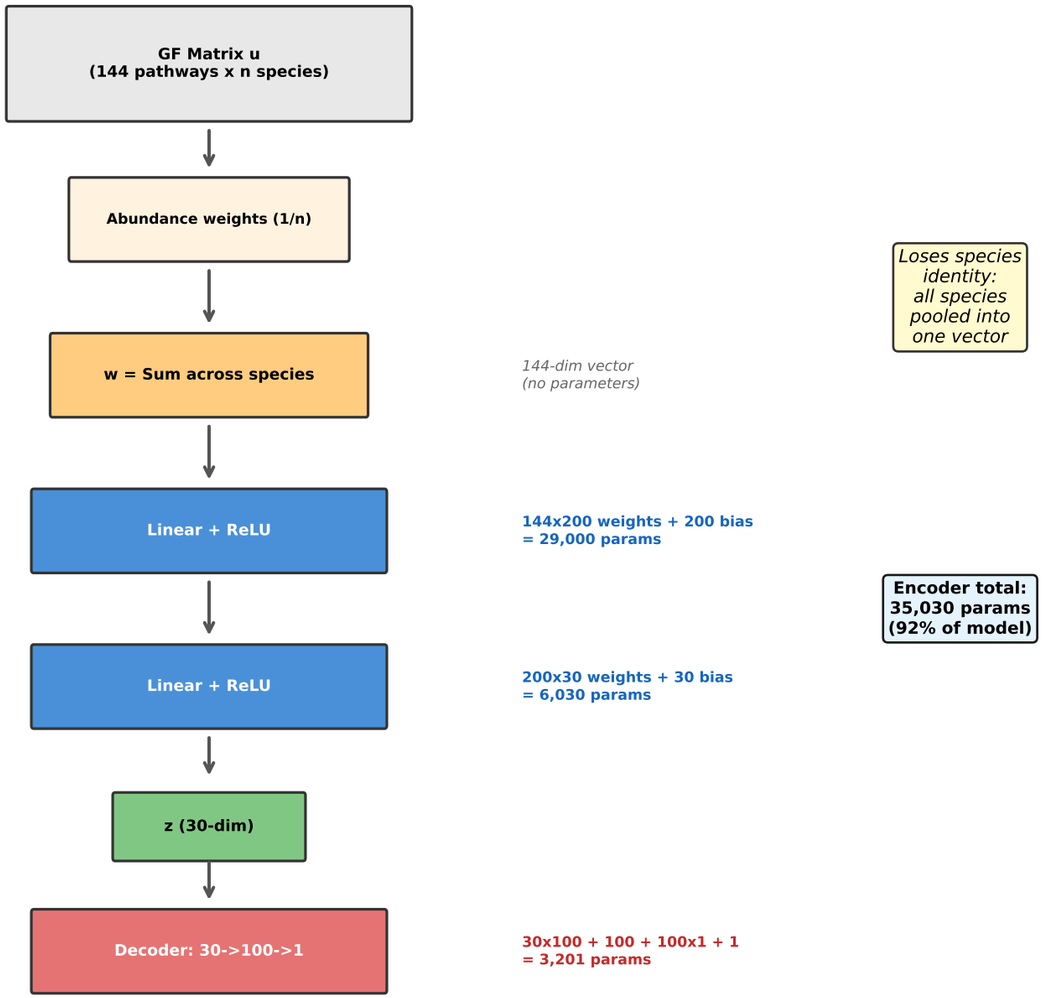
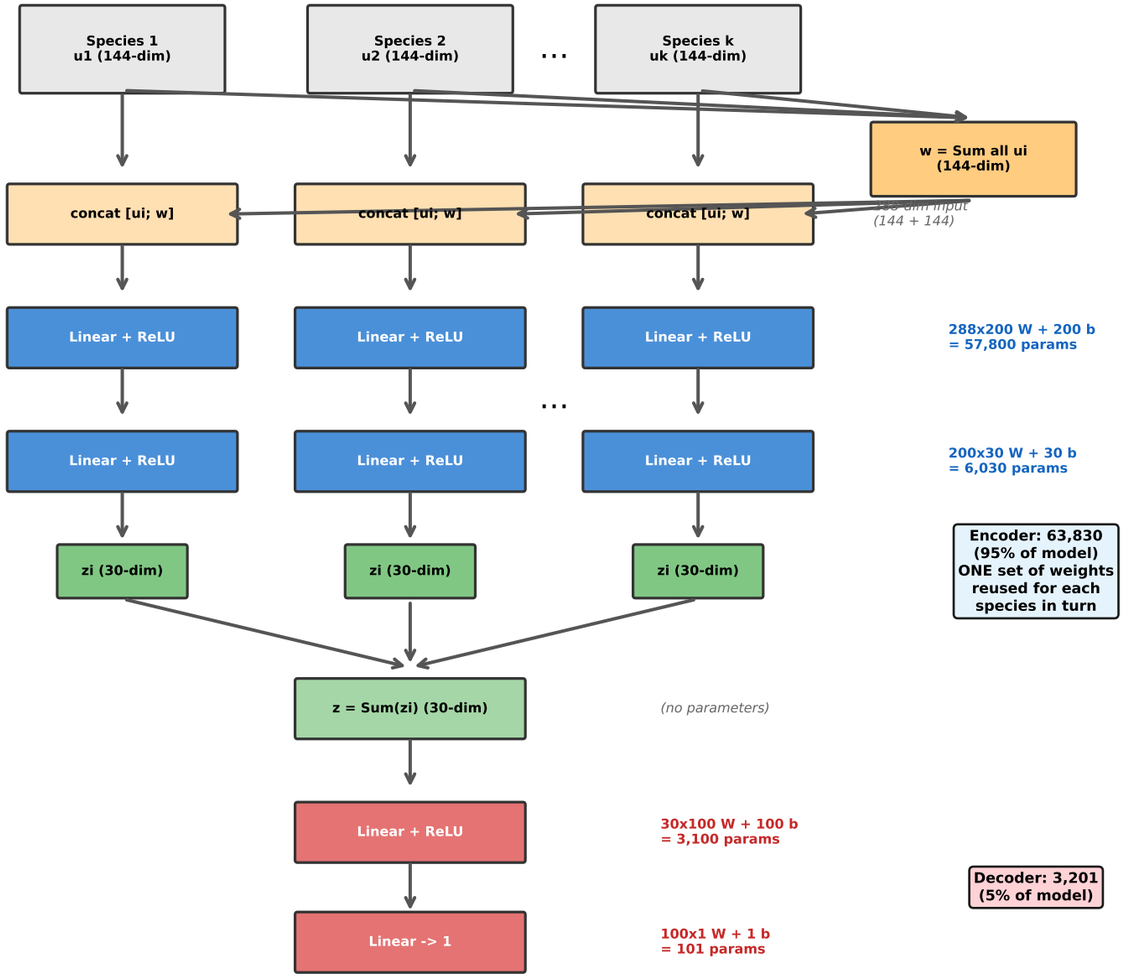
Candidate realizations of dCGF and their performances on synthetic data. (A) A common model structure of candidate dCGFs. The community GF matrix u is first fed into an encoder $z = \alpha(u)$ to extract the community embedding z. The community embedding z is then fed into the decoder $y = \sigma(z)$, which is a fully connected neural network to predict community function(s). The two candidate realizations of dCGFs differ in their encoder structures. (B) The encoder of a dCGF-enzyme soup (dCGF-ES) model sums up the GF vectors of all species to form a community GF vector w, which is then fed into a fully connected neural network to output the community embedding z. (C) By contrast, the encoder of a dCGF-individual species (dCGF-IS) model process the genetic feature vector of each species individually. Specifically, the abundance-weighted GF vector of each species i, u_i, and the community gene pool vector, w, are fed into a neural network to obtain each species' functional embedding z_i. The functional embeddings are summed up to become the community embedding z. The neural networks processing each species' GF vector have identical weights and structures. (D) dCGF-IS is strictly more expressive than dCGF-ES. For every species genotype to community function mapping represented by dCGF-ES (buff-colored set), there exists a parametrization of dCGF-IS (gray-colored set) that can represent the same genotype-function mapping. A subset of community genotype-function mappings (pink-colored set in panel D) can be represented by dCGF-IS but cannot be represented by dCGF-ES. (E) An example of species genotype to community function mapping that cannot be represented by dCGF-ES. The two communities have different GF matrices, resulting in different output metabolite productions (high vs. low). However, dCGF-ES will make same predictions for the two communities since the column-sum of the two communities' GF matrices are identical. (F) Mechanistic microbe-effector models (MEM) were used to generate synthetic data to train/test two dCGF realizations and compare their performances. Specifically, training and testing species were each assigned a random GF vector, which was then used to determine its MEM parameters, such as resource consumption rates and toxin production rates. Simulated function of MEMs from training communities were used to train dCGFs to predict the functions of testing communities. All species in testing communities do not appear in training data. The MEM model includes multiple modes of inter-species interactions and intra-cellular regulations. See **Methods** for more details on synthetic data generation. (F) Testing performances of the two different dCGF realizations for different training epochs. Testing performances were evaluated as the Pearson correlation between community functions simulated by ground-truth MEM and predicted by dCGF. Shaded areas quantify 95% confidence intervals arising from ten different random seeds used to generate the MEM parameters as well as to initialize neural network weights in dCGF (see **Methods**).

Model Architectures: dCGF-ES vs dCGF-IS
ES pools all species into one vector; IS (Deep Set) preserves per-species processing

dCGF-ES (Enzyme Soup) — 38,231 total parameters

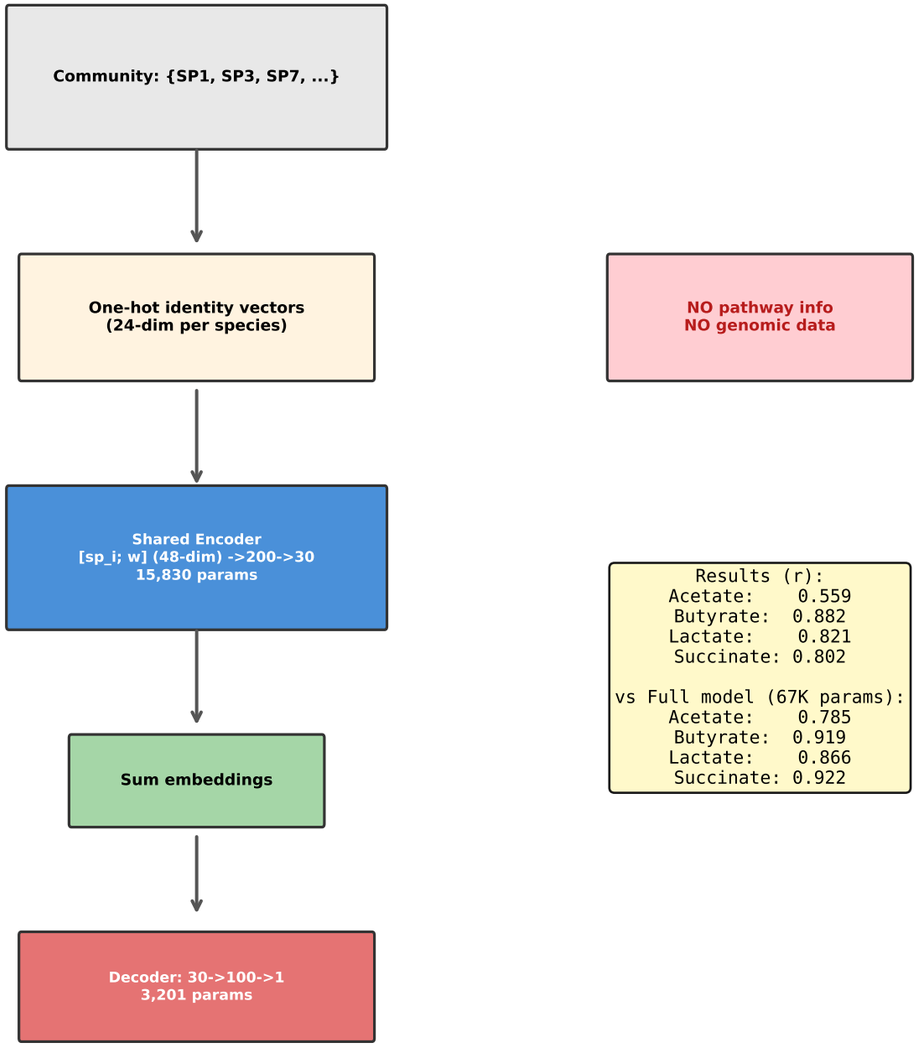


dCGF-IS (Individual Species) — Deep Set — 67,031 total parameters



Control Models: Ablating Pathway Information

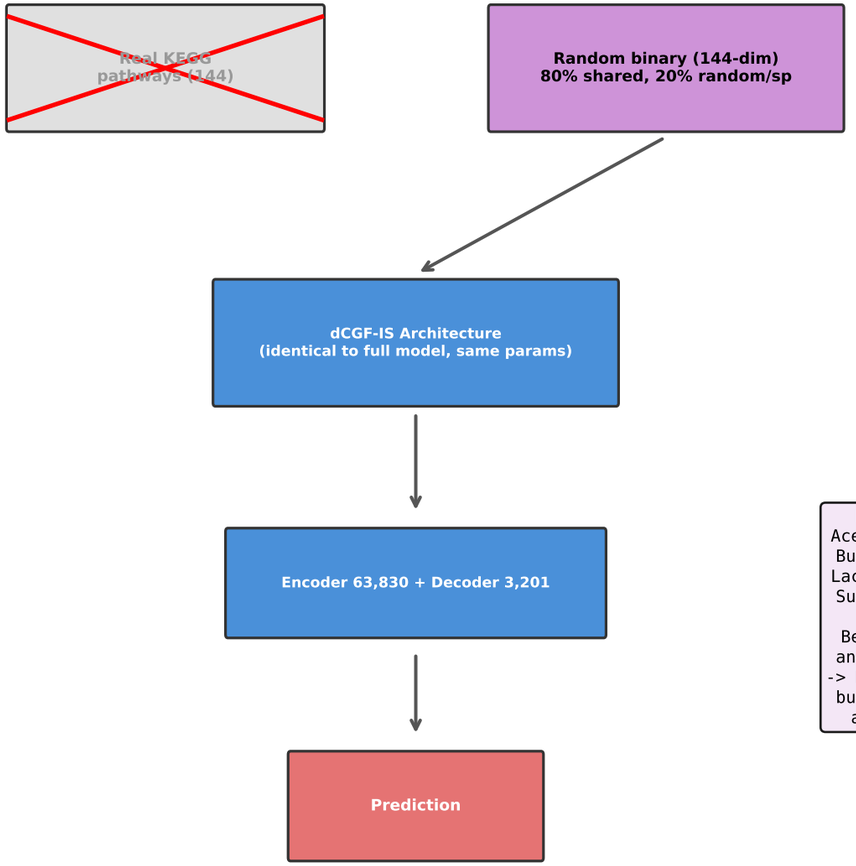
Species-Only Model — 19,031 total parameters



NO pathway info
NO genomic data

```
Results (r):
Acetate:    0.559
Butyrate:   0.882
Lactate:    0.821
Succinate:  0.802

vs Full model (67K params):
Acetate:    0.785
Butyrate:   0.919
Lactate:    0.866
Succinate:  0.922
```

Random Pathways Control — 67,031 total parameters




```
Results (r):
Acetate:    0.702
Butyrate:   0.892
Lactate:    0.854
Succinate:  0.874

Between Real PW
and Species-Only
-> Extra dims help
but real biology
adds modestly
```

Simple Baselines: Comparison with Traditional Models

Ridge Regression (Linear)

Species presence/absence (binary vector)

(24 features)

↓

Ridge Regression (L2-regularized linear model)

↓

Prediction

```
Results (r):
Acetate    0.591
Butyrate   0.867
Lactate    0.772
Succinate  0.692
```

*Linear: y = w1*sp1 + w2*sp2 + ... + b*

Random Forest (Species Only)

Species presence/absence (binary vector)

(24 features)

↓

Random Forest (100 trees, nonlinear)

↓

Prediction

```
Results (r):
Acetate    0.537
Butyrate   0.893
Lactate    0.824
Succinate  0.809
```

Captures nonlinear species interactions

Random Forest (Species + Pathways)

Species presence (24-dim) + Summed pathway vectors (144-dim)

(168 features)

↓

Random Forest (100 trees, nonlinear)

↓

Prediction

```
Results (r):
Acetate    0.677
Butyrate   0.905
Lactate    0.840
Succinate  0.877
```

Best RF: pathways add real signal over species alone

Random Forest (Species + Pairwise Similarity)

Species presence (24-dim) + Pairwise pathway correlations

(24 + 276 = 300 features)

↓

Random Forest (100 trees, nonlinear)

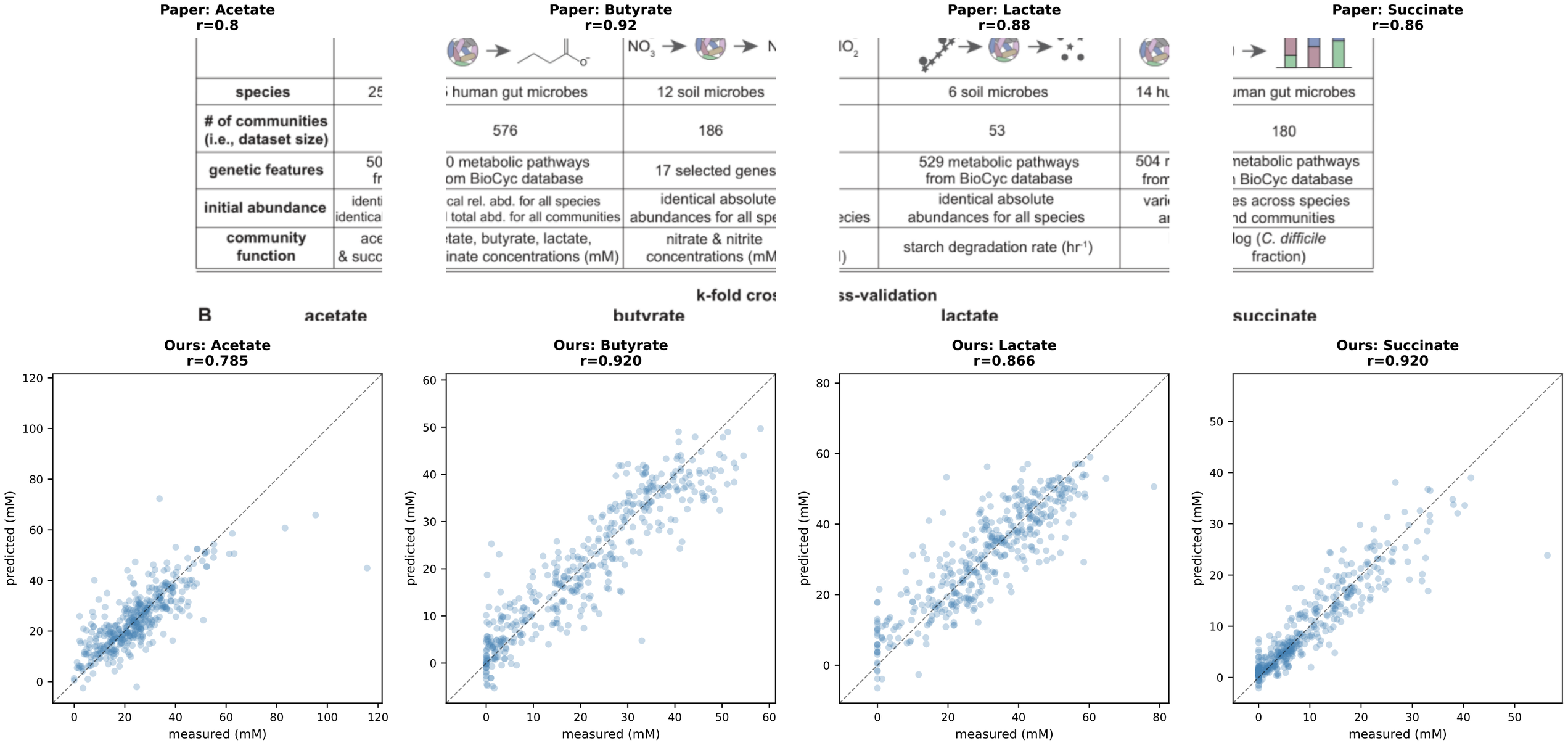
↓

Prediction

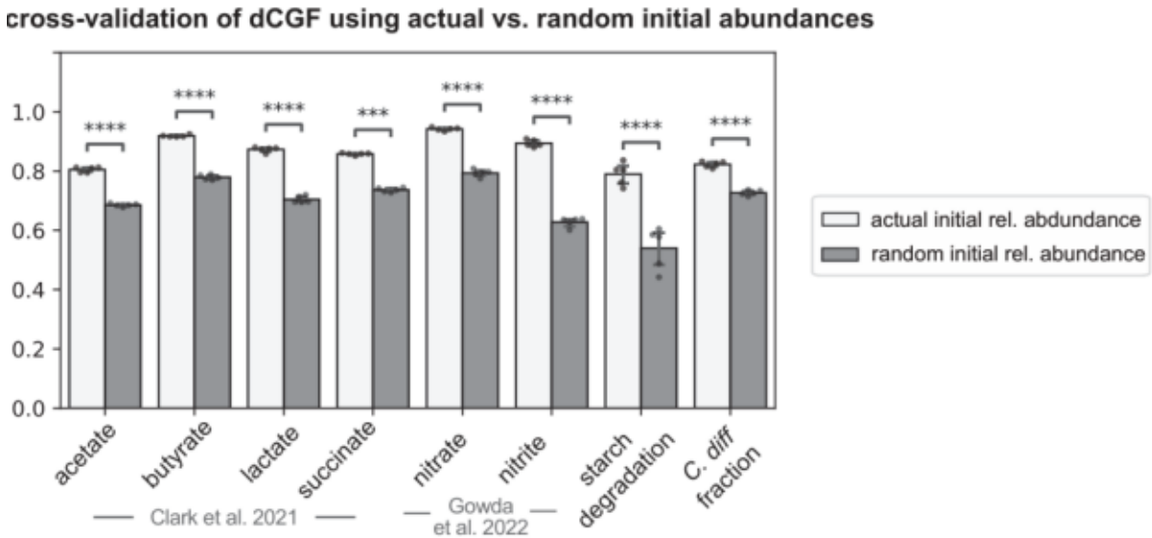
```
Results (r):
Acetate    0.519
Butyrate   0.891
Lactate    0.819
Succinate  0.790
```

Similarity does not improve over species-only

Figure 3B Reproduction: Predicted vs Measured Metabolite Concentrations
Top: Paper (BioCyc, 500 pathways, 25 species) | Bottom: Ours (KEGG, 144 pathways, 23 species)

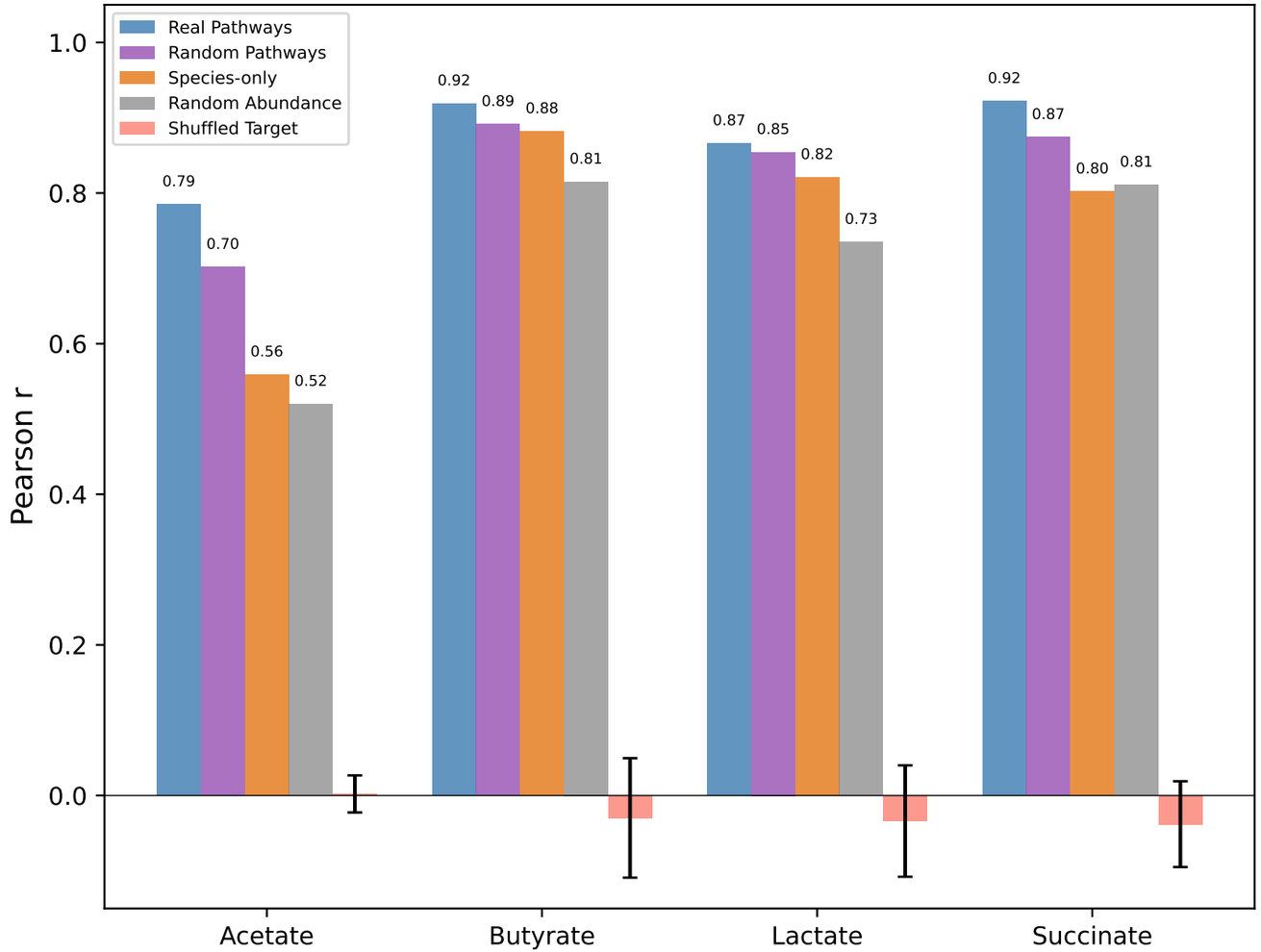


Paper Figure 3C: Actual vs Random Abundance
(BioCyc, 500 pathways, 25 species)

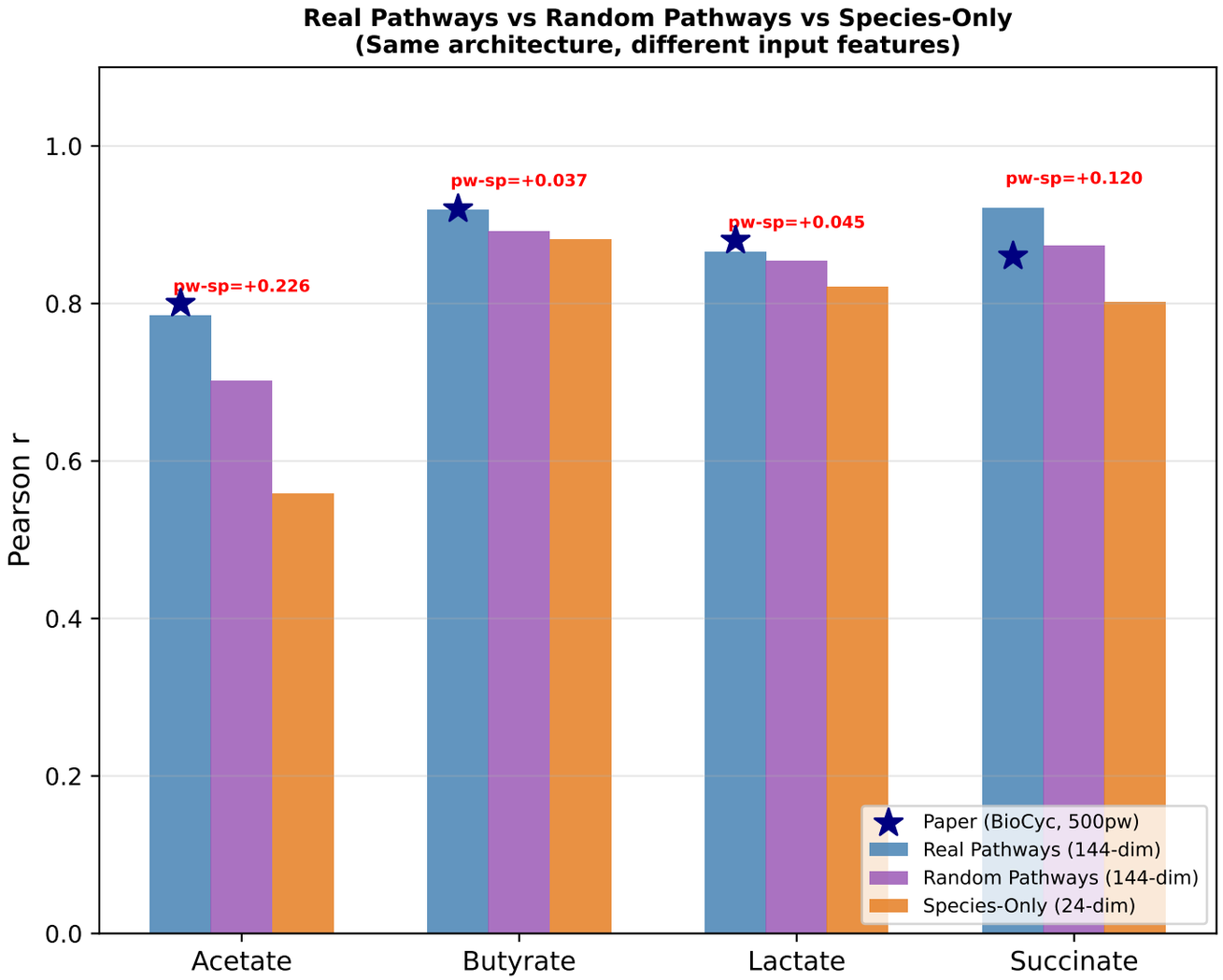


dCGF performance on experimental data using k-fold cross-validation. (A
xperimental datasets [27], [36], [46], [47] used to train and test dCGF-IS. (B) Scatter p
asured community functions for all experimental datasets. Five different random se
k folds and to initialize neural network parameters. Predicted function of a communit
ctions. (See **Methods**.) (C) Bar plot of k-fold cross-validation performances of dCGF

Our Results: All Conditions
(500 epochs, 5 seeds, 8-fold CV)

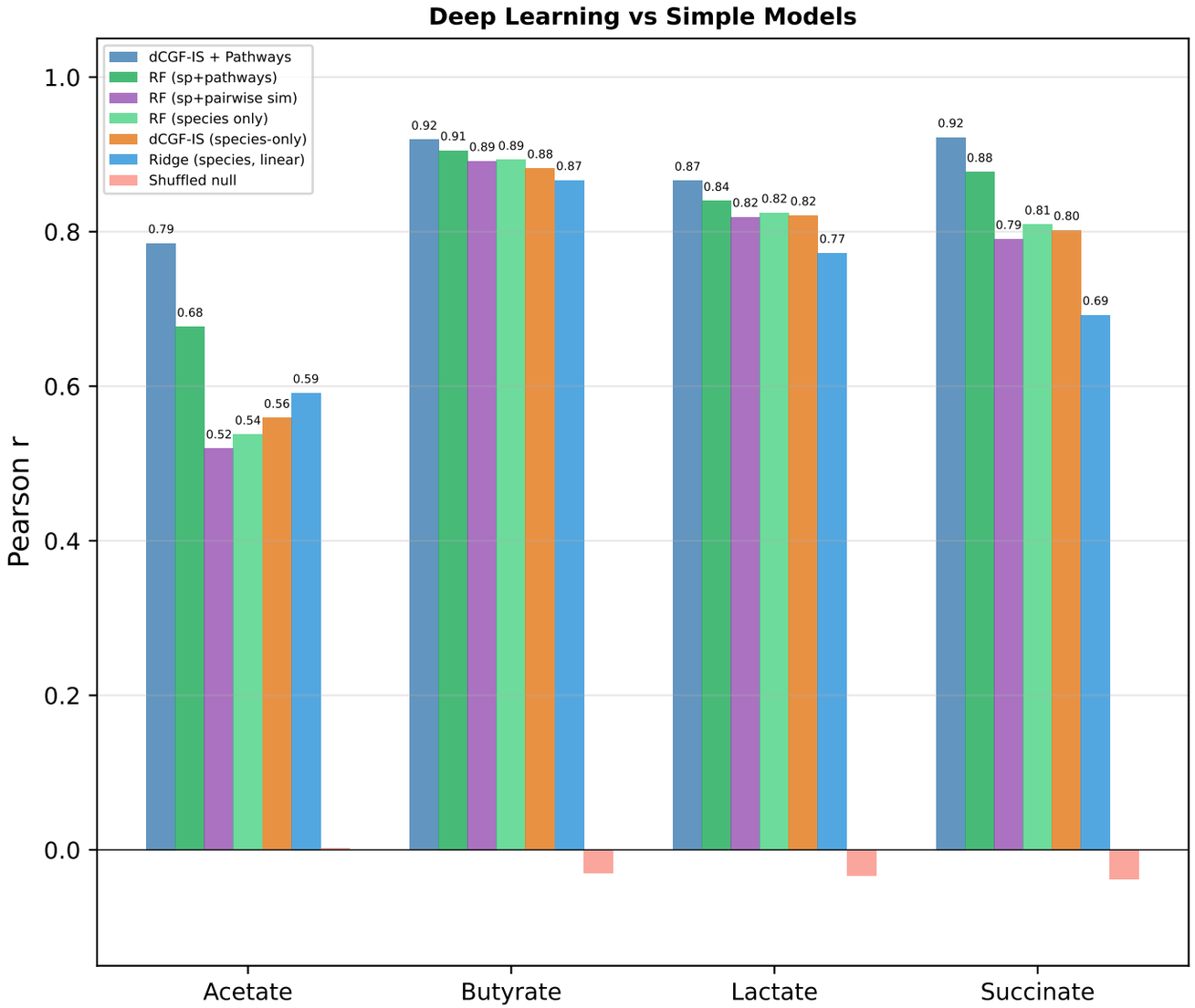


Species Identity vs. Pathway Information



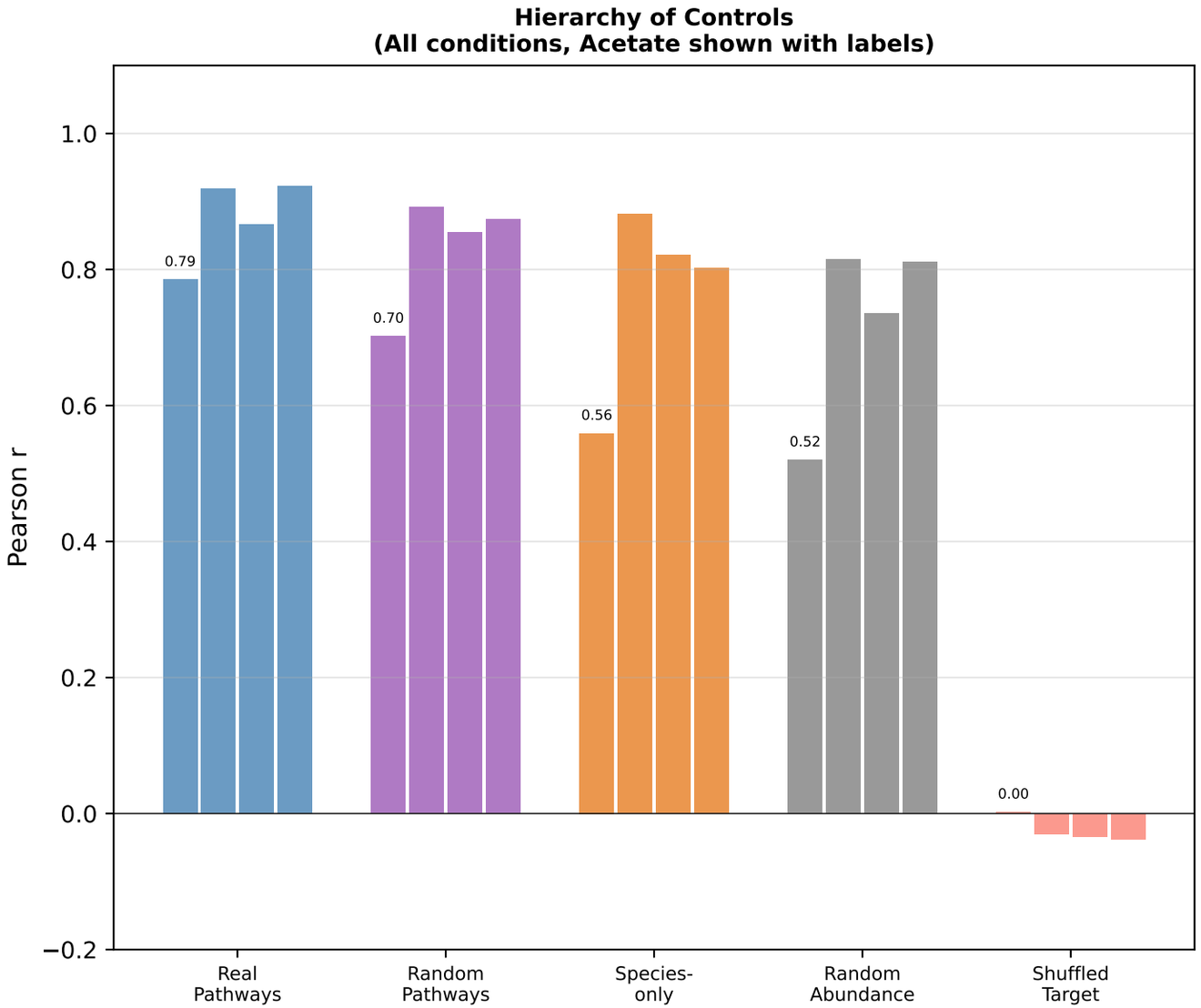

```
EXPERIMENTAL DESIGN
-------------------
Real pathways: species' actual 144-dim KEGG vector
Random pathways: 144-dim, 80% shared across all
  species, 20% randomly assigned (no biology)
Species-only: 24-dim one-hot identity vector
  (NO genomic/pathway info at all)

Same dCGF-IS architecture, same training,
same k-fold CV, same communities.

RESULTS
-------
Metabolite  Real-PW  Rand-PW  Sp-Only
----------  -------  -------  -------
Acetate     0.785    0.702    0.559
Butyrate    0.919    0.891    0.882
Lactate     0.866    0.854    0.821
Succinate   0.922    0.874    0.802

INTERPRETATION
--------------
If Random Pathways ~ Species-Only:
  Extra dimensions don't help; model uses
  species identity from one-hot or pathway.

If Random Pathways ~ Real Pathways:
  Any distinguishing features work; real
  pathway biology is irrelevant.

If Real Pathways >> Random Pathways:
  Pathway content genuinely matters for
  those metabolites.
```

Simple Baselines: Random Forest and Ridge Regression

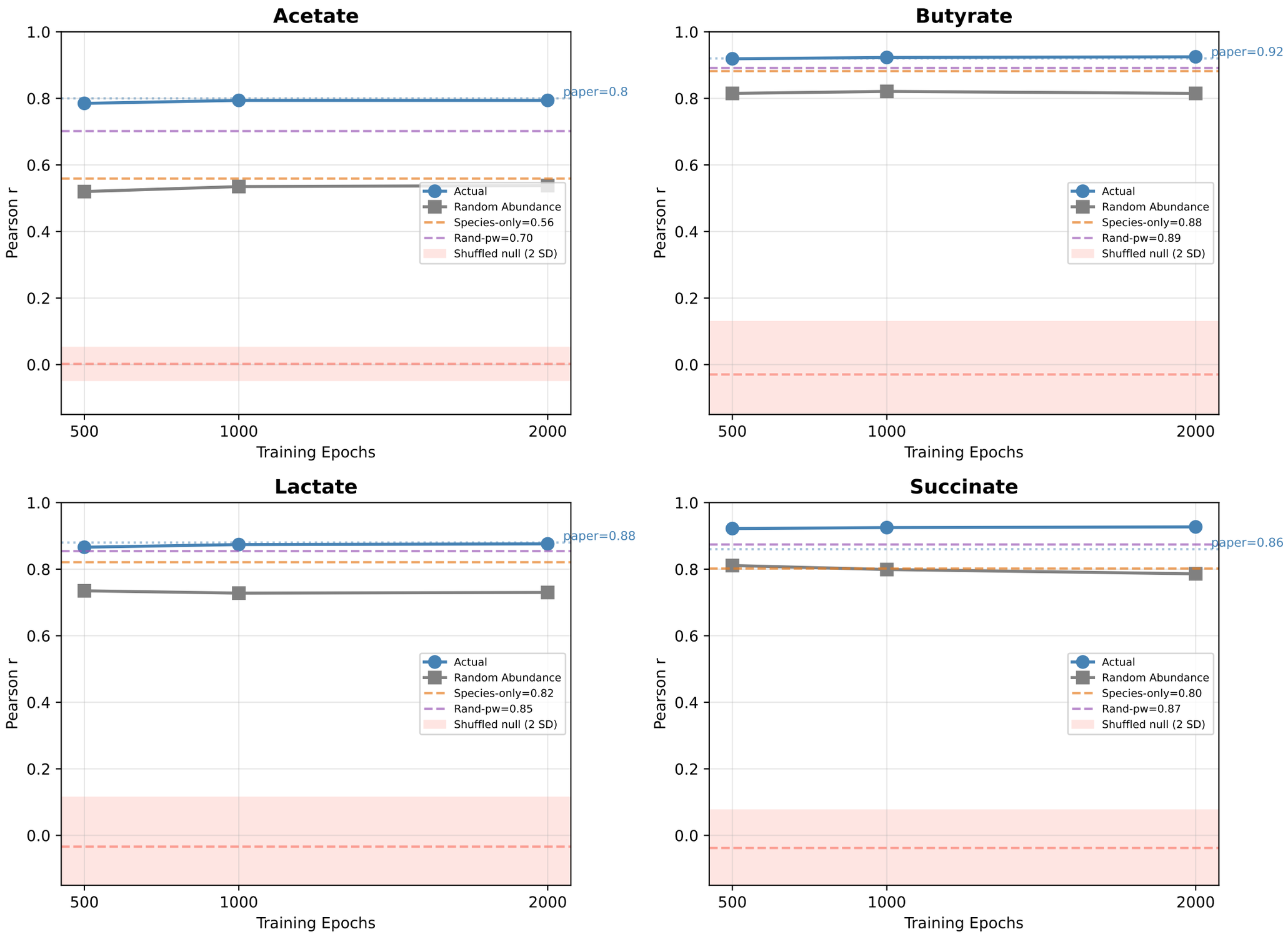


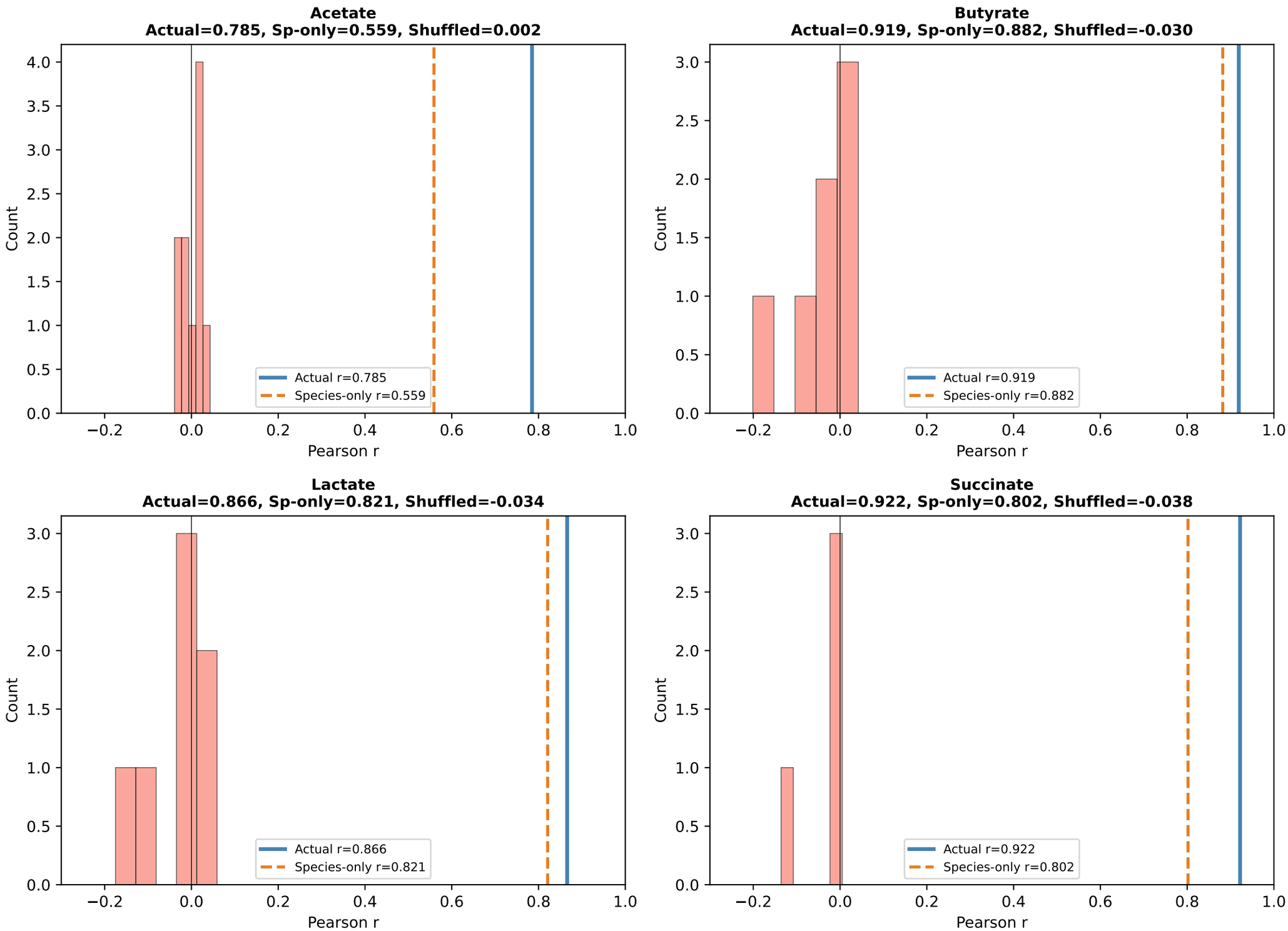

```
DEEP LEARNING vs TRADITIONAL MODELS
===================================

RESULTS
-------
Metab   dCGF RF+PW RF+Sim RF(sp) Ridge
------- ---- ----- ------ ------ -----
Acetate 0.785 0.677 0.519  0.537  0.591
Butyrate0.919 0.905 0.891  0.893  0.867
Lactate 0.866 0.840 0.819  0.824  0.772
Succinate0.922 0.877 0.790  0.809  0.692

KEY OBSERVATIONS
----------------
1. RF(sp+pw) improves over RF(sp) for all
   metabolites -> pathways add real signal

2. RF(sp+sim) does not improve over
   species-only -> pathway value is in
   specific content, not genomic similarity

3. Specific pathway content matters, not
   just how genomically similar co-occurring
   species are to each other

4. dCGF-IS still beats RF(sp+pw) everywhere
   -> Deep Set architecture extracts more
   from per-species pathway vectors

5. Ridge (linear) close for Butyrate (0.87)
   -> species effects largely additive
```

Understanding the Random-Abundance Control




```
UNDERSTANDING THE RANDOM ABUNDANCE CONTROL
==========================================

The paper's negative control randomizes species
abundances while keeping pathway data intact.

Context: Clark et al. used equal abundances
(1/n for all species), so the actual abundances
are uniform. Randomizing them perturbs a uniform
distribution, limiting the information removed.

Additionally, binary pathway features provide
near-unique species fingerprints (100-120 of 144
pathways per species). With random weights, the
model can still infer species identity from
pathway patterns.

COMPLEMENTARY CONTROLS:
-----------------------
1. Shuffled-target permutation
   -> r ~ 0: confirms real signal exists

2. Species-only model
   -> r = 0.56-0.88: species composition is
   the dominant predictor

3. Random-pathway assignment
   -> Tests if pathway structure matters
   vs. just having distinguishing features
```


Epoch Progression: Performance Plateaus at 500 Epochs
Orange dashed = species-only baseline (no pathway info needed)
Acetate
Butyrate
Lactate
Succinate
Pearson r
Training Epochs
500
1000
2000
1.0
0.8
0.6
0.4
0.2
0.0
paper=0.8
paper=0.92
paper=0.88
paper=0.86
Actual
Random Abundance
Species-only=0.56
Rand-pw=0.70
Species-only=0.88
Rand-pw=0.89
Species-only=0.82
Rand-pw=0.85
Species-only=0.80
Rand-pw=0.87
Shuffled null (2 SD)


Permutation Test: Shuffled Target Null Distribution
Both pathway and species-only models far exceed shuffled null
Acetate
Actual=0.785, Sp-only=0.559, Shuffled=0.002
Actual r=0.785
Species-only r=0.559
Butyrate
Actual=0.919, Sp-only=0.882, Shuffled=-0.030
Actual r=0.919
Species-only r=0.882
Lactate
Actual=0.866, Sp-only=0.821, Shuffled=-0.034
Actual r=0.866
Species-only r=0.821
Succinate
Actual=0.922, Sp-only=0.802, Shuffled=-0.038
Actual r=0.922
Species-only r=0.802
Count
Pearson r

Summary: All Conditions (5 seeds, 8-fold CV, 500 epochs)

Metabolite	Paper (BioCyc)	dCGF-IS (KEGG)	dCGF-IS (fake PW)	RF (sp+pw)	RF (sp+sim)	RF (species)	dCGF-IS (sp-only)	Ridge (species)	dCGF-IS (shuf wt)	Shuffled Targets	Paper Rand Wt
Acetate	0.80	0.785	0.702	0.677	0.519	0.537	0.559	0.591	0.520	0.002	~0.70
Butyrate	0.92	0.919	0.891	0.905	0.891	0.893	0.882	0.867	0.815	-0.030	~0.78
Lactate	0.88	0.866	0.854	0.840	0.819	0.824	0.821	0.772	0.735	-0.034	~0.75
Succinate	0.86	0.922	0.874	0.877	0.790	0.809	0.802	0.692	0.811	-0.038	~0.82

See next page for column definitions and key observations.

Summary Table: Column Definitions & Key Observations

```
Column definitions:

  dCGF-IS (KEGG)     = Full model: real species + real KEGG pathways (144-dim)
  dCGF-IS (fake PW)  = Real species, but pathways replaced with random binary vectors
  RF (sp+pw)         = Random Forest on species presence + summed pathway vectors (168-dim)
  RF (sp+sim)        = Random Forest on species + pairwise pathway correlations (300-dim)
                       Tests if phylogenetic similarity explains pathway value
  RF (species)       = Random Forest on species presence/absence only (24-dim)
  dCGF-IS (sp-only)  = dCGF-IS with one-hot species identity, NO pathway info
  Ridge (species)    = Linear regression on species presence/absence (24-dim)
  dCGF-IS (shuf wt)  = Real pathways, but species weights randomized (paper's control)
  Shuffled Targets   = Real inputs, but metabolite labels permuted (proper null)
  Paper Rand Wt      = Paper's reported random-weight control (BioCyc, from Fig 3C)
```

```
Key observations:

  1. Species-only models (RF, Ridge, dCGF) get r = 0.54-0.89 with NO pathway data
  2. RF(sp+pw) improves over RF(sp), but dCGF-IS still wins -> architecture matters
  3. RF(sp+sim) does not improve -- phylogenetic similarity is not what pathways encode
  4. Specific pathway content matters, not just genomic similarity between species
  5. Shuffled targets yield r ~ 0 (standard permutation null)
```

dCGF-ES (Enzyme Soup) vs dCGF-IS (Individual Species)

Metabolite	dCGF-ES (ours)	dCGF-IS (ours)	IS - ES (gain)	Paper IS (BioCyc)
Acetate	0.705 +/- 0.076	0.785 +/- 0.058	+0.080	0.80
Butyrate	0.903 +/- 0.003	0.919 +/- 0.004	+0.016	0.92
Lactate	0.814 +/- 0.006	0.866 +/- 0.011	+0.052	0.88
Succinate	0.850 +/- 0.010	0.922 +/- 0.005	+0.072	0.86

dCGF-ES (Enzyme Soup) sums all species' pathway vectors into one community vector before encoding — losing per-species identity. dCGF-IS (Individual Species) encodes each species separately with a shared encoder, preserving per-species information.

IS consistently outperforms ES, especially for Succinate (+0.071) and Acetate (+0.080). This confirms that per-species processing matters — the model benefits from knowing WHICH species carry which pathways, not just the total pathway count.

Note: The paper does NOT report dCGF-ES results on the Clark et al. dataset. They only compare ES vs IS on synthetic data (Figure 2E), where IS also outperforms ES. Our results are the first ES vs IS comparison on real experimental data with KEGG pathways.

Interpretability Methods: Paper's Sensitivity Analysis

```
THE PAPER'S METHOD (Qian et al. Figure 4)
=========================================

Goal: Determine which pathways matter most for predicting
each metabolite.

Procedure:
  1. Train dCGF-IS on ALL available data
  2. Construct a 'nominal' GF matrix containing ALL 25 species
     (a hypothetical community with every species present)
  3. For each pathway p in each species s:
     a. If pathway is PRESENT (=1): silence it (set to 0)
     b. If pathway is ABSENT (=0): activate it (set to 1)
     c. Run the perturbed matrix through the trained model
     d. Compute: sensitivity = (F(perturbed) - F(nominal)) / delta
        where delta = change in the pathway value
  4. Average |sensitivity| across all species and random seeds
  5. Rank pathways by average |sensitivity|

Interpretation:
  - High |sensitivity| = toggling this pathway changes
    the predicted metabolite concentration a lot
  - Sign indicates direction: positive = enhances production,
    negative = inhibits production

IMPORTANT CONSIDERATION:
  If a pathway is present in only 1 species (e.g., Siderophore
  biosynthesis in AC only), silencing it is equivalent to
  partially removing that species from the community.
  In such cases, sensitivity may reflect species importance
  rather than pathway importance. Pathways shared by 24/25
  species (e.g., Glycolysis, TCA cycle) show lower sensitivity
  because toggling them in one species has less community impact.
```

Our Interpretability Methods: Permutation Importance & Sensitivity

```
METHOD 1: PERMUTATION IMPORTANCE (model-agnostic)
=================================================

Goal: Which pathways carry the most predictive information?

Procedure:
  1. Train an ensemble of 5 dCGF-IS models (different seeds)
  2. Compute baseline Pearson r using ensemble-averaged predictions
  3. For each of 144 pathways:
     a. SHUFFLE that pathway's values across species WITHIN each
        community (breaks the species-pathway association while
        preserving marginal distributions)
     b. Re-predict with shuffled data
     c. importance = baseline_r - shuffled_r
     d. Repeat 10 times, take mean
  4. Rank pathways by importance (drop in r)

Key difference from paper's method:
  - Permutation importance does NOT add/remove pathways --
    it scrambles them, destroying the species-pathway link
  - Measures information content, not causal effect
  - Less confounded by pathway prevalence


METHOD 2: SENSITIVITY ANALYSIS (reproduction of paper's method)
===============================================================

We reproduce the paper's Figure 4 approach using our KEGG data:
  - For each pathway in each species in each community:
    silence (1->0) or activate (0->1) and measure prediction change
  - Average |sensitivity| across species, communities, and seeds
  - This allows direct comparison with the paper's results

The following pages show BOTH methods side-by-side for each
metabolite, followed by a prevalence analysis
(examining the relationship between sensitivity and rarity).
```

Paper Figure 4: Sensitivity Analysis
Authors silence/activate each BioCyc pathway to measure impact on predictions

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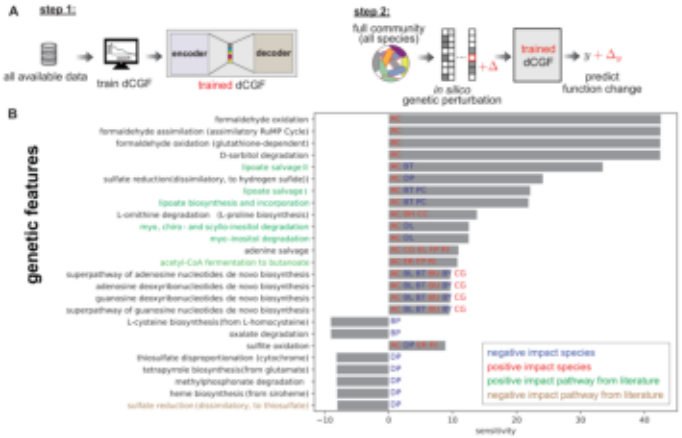
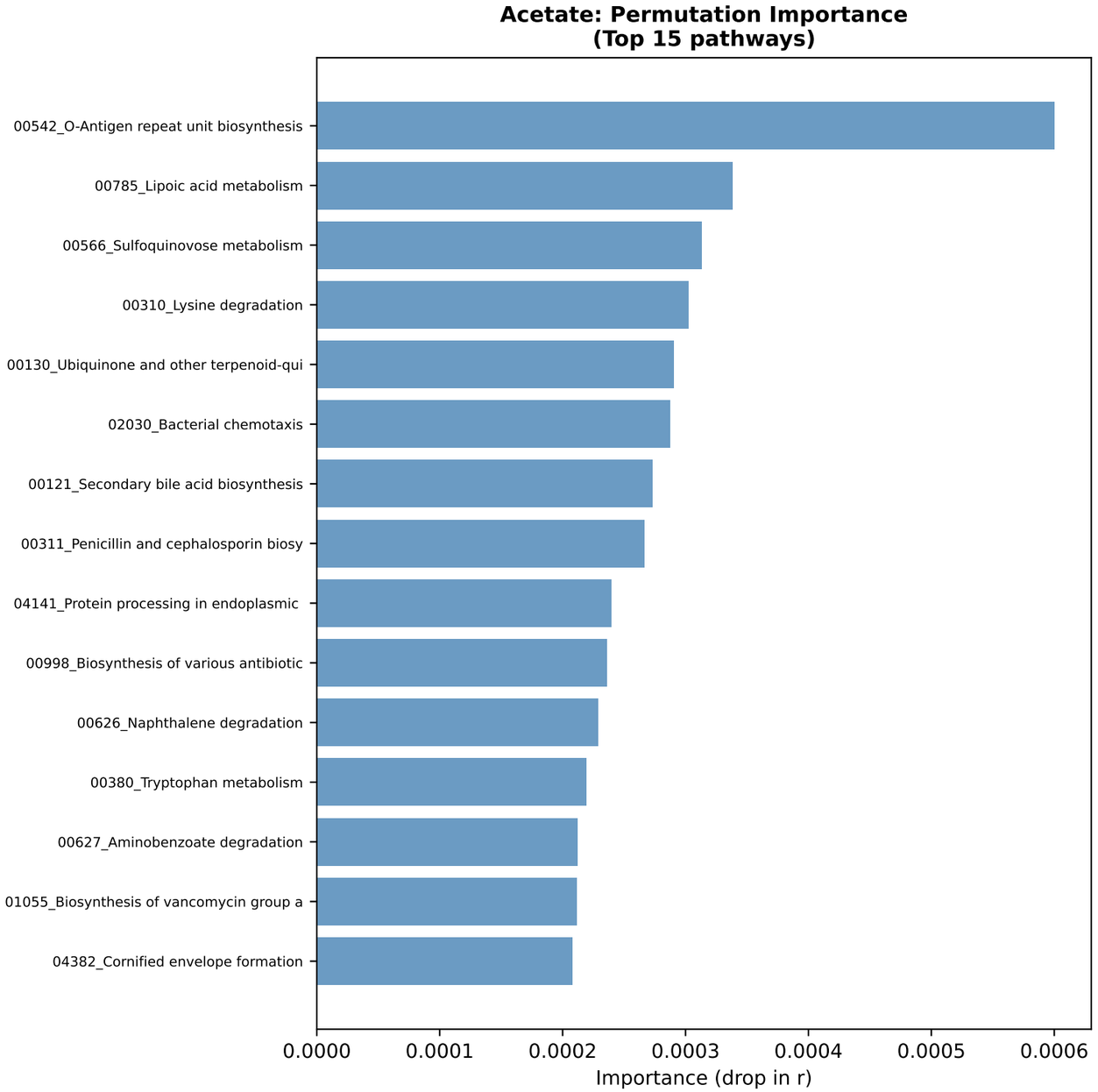
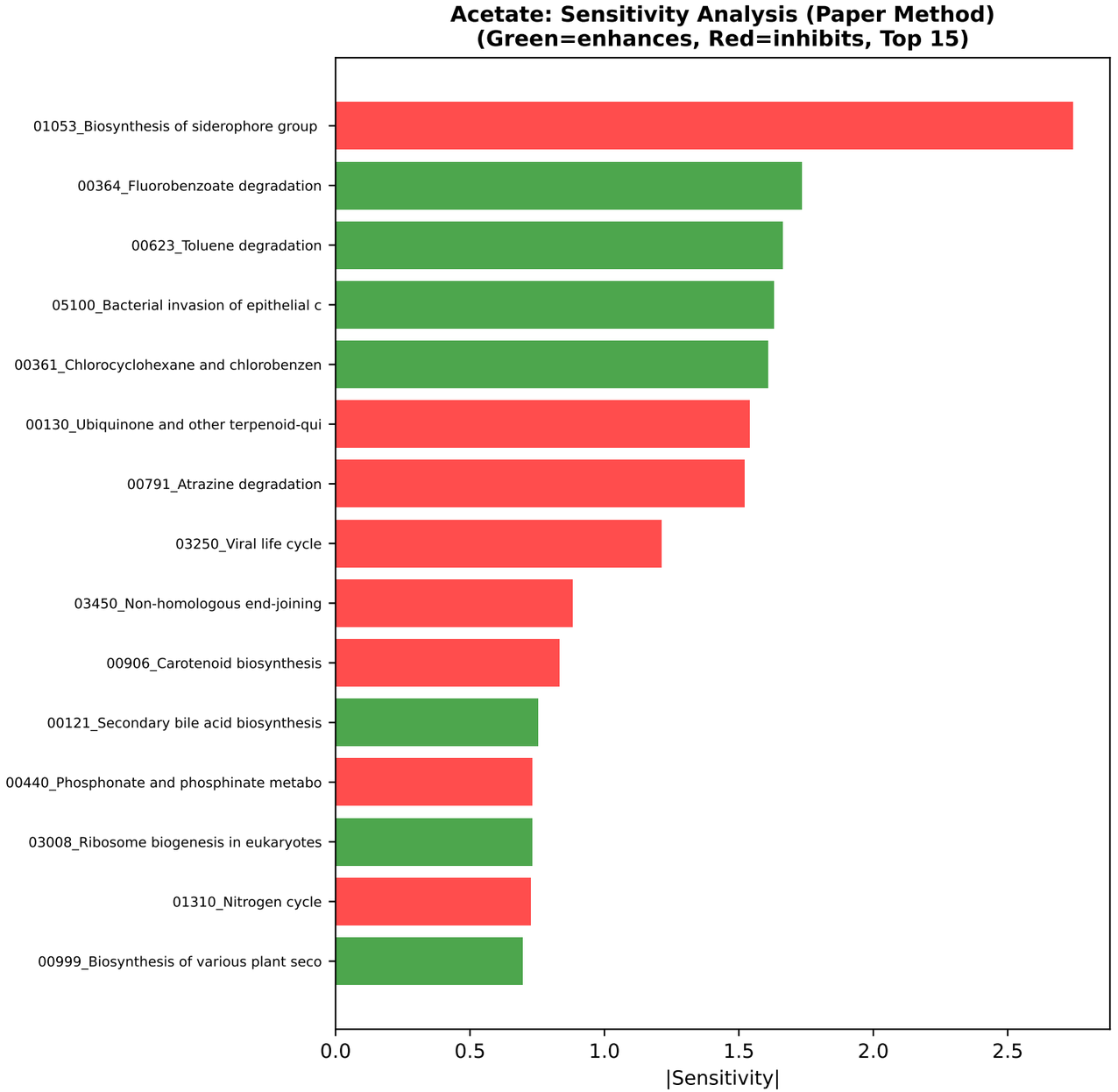
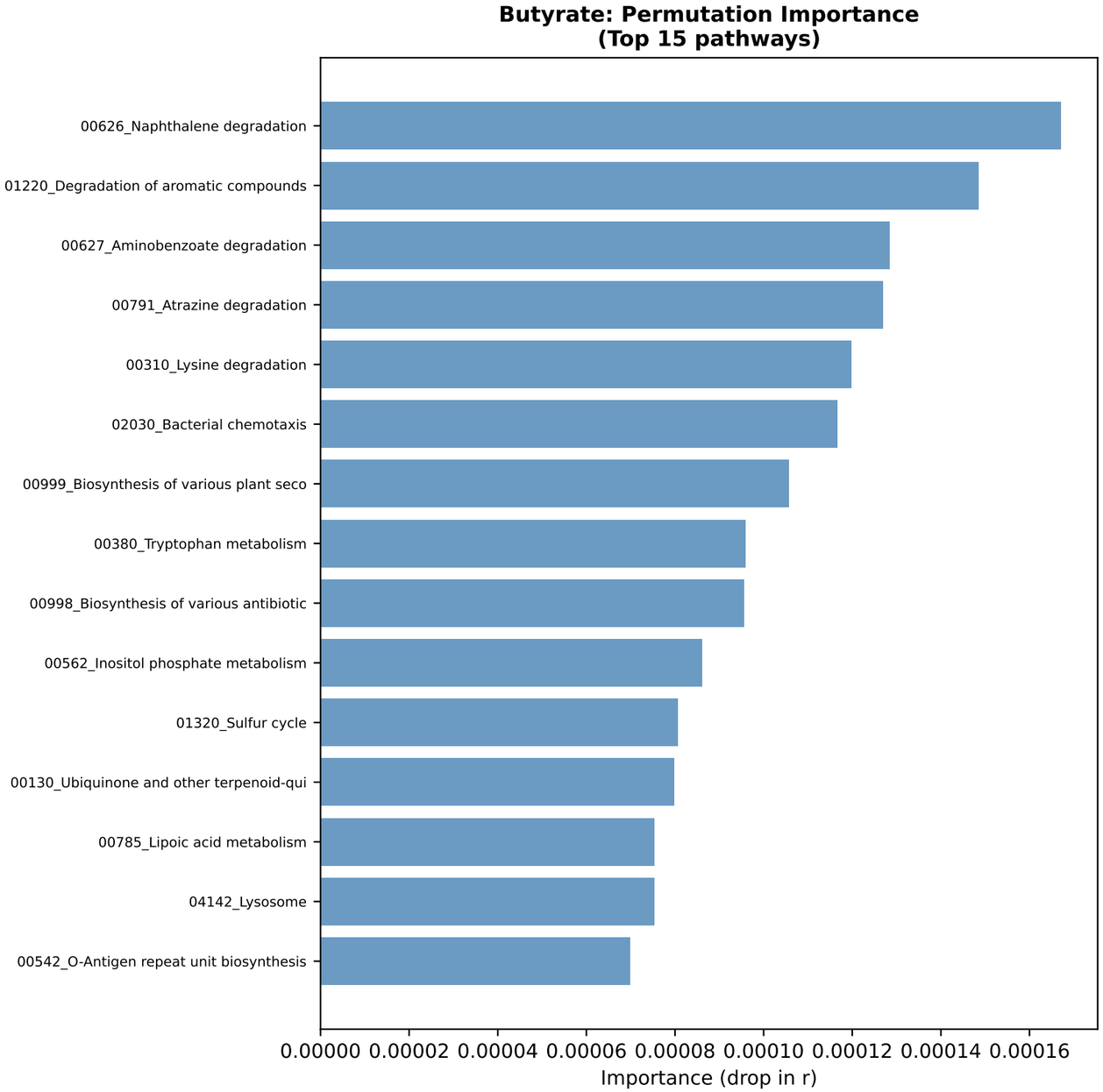
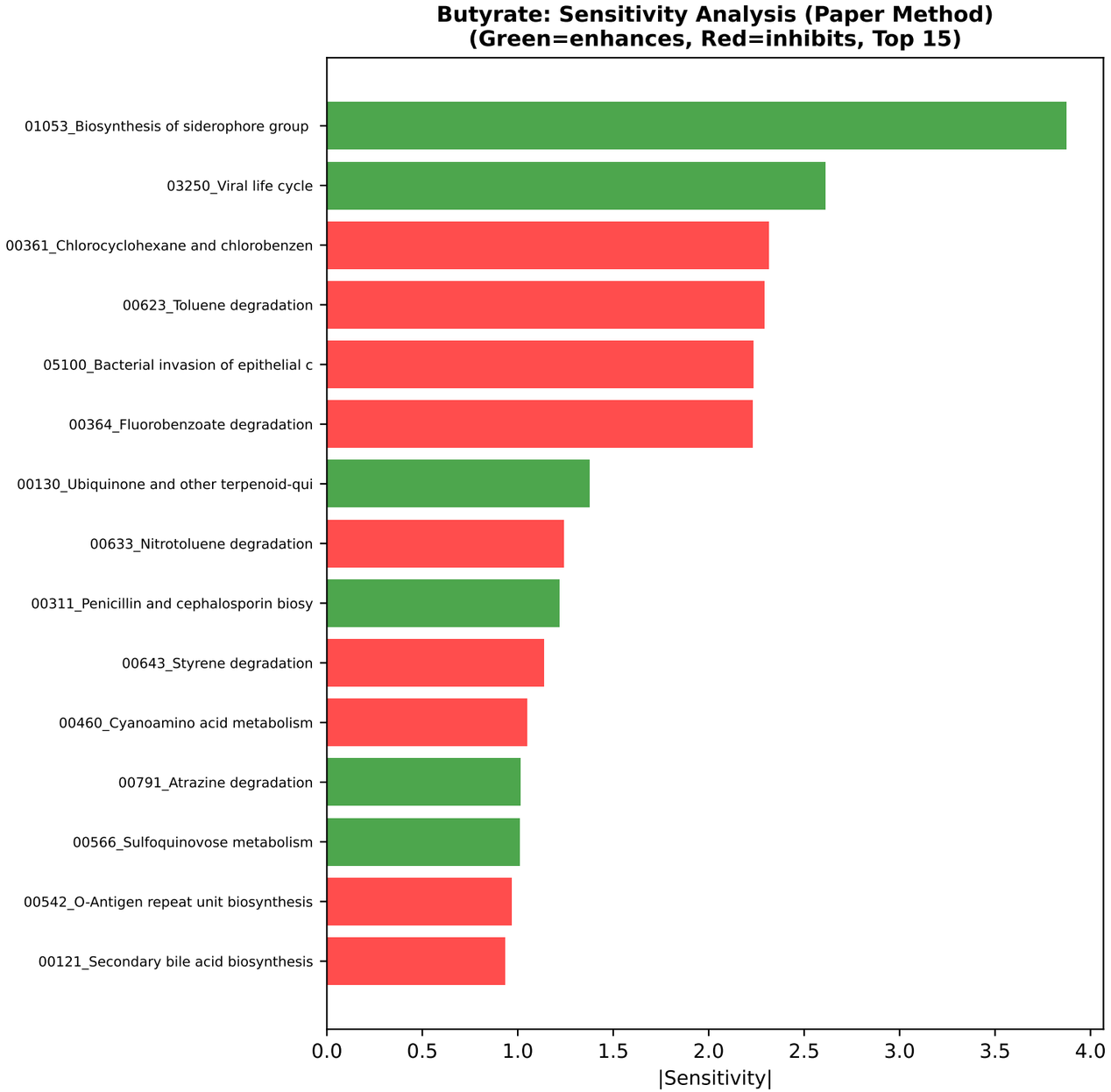


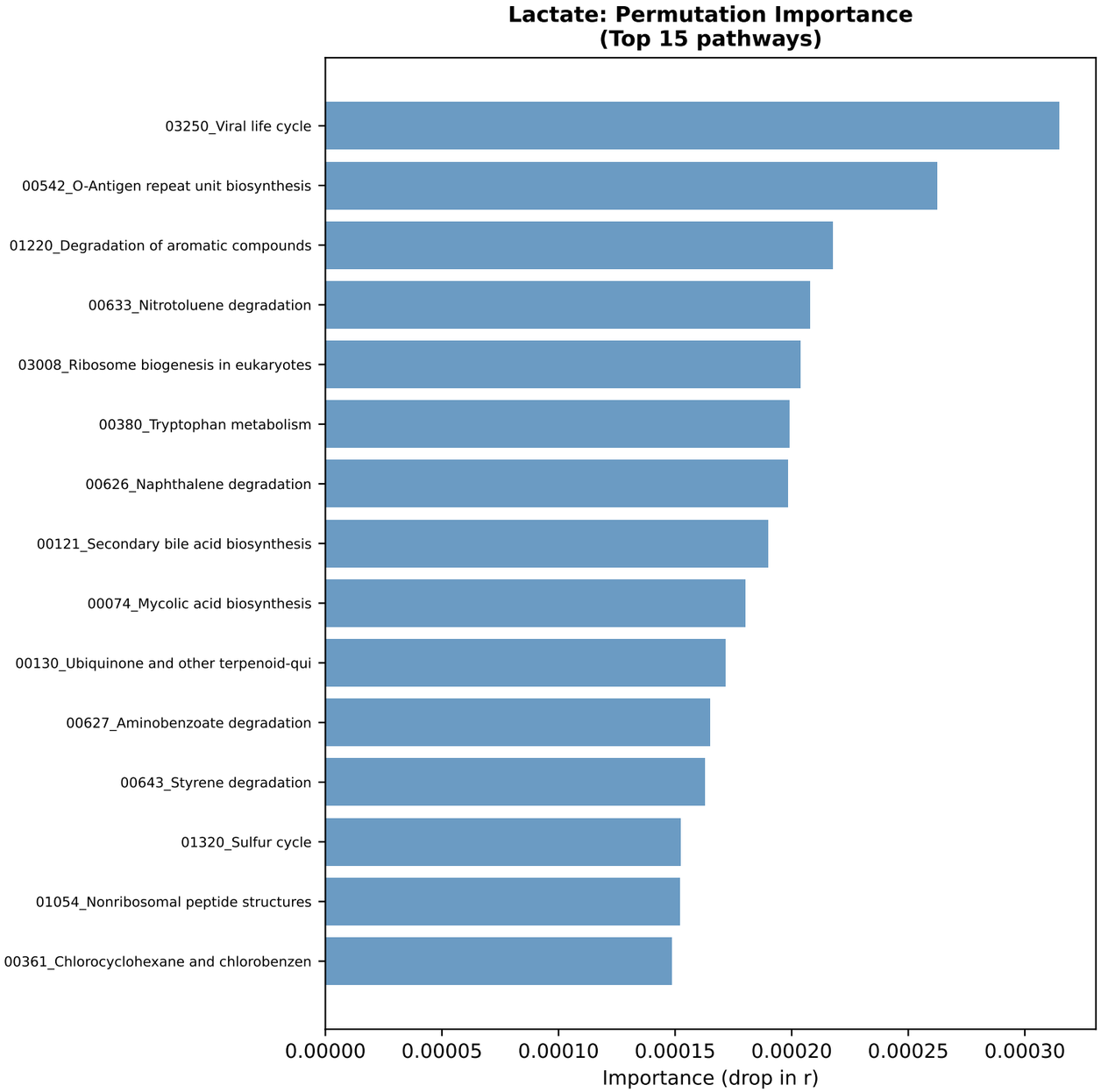
Figure 4: **Sensitivity analysis of dCGF to evaluate the impact of each GF on community function.** (A) Workflow to compute sensitivities of each GF. All available data for a community function was first used to train a dCGF-IS. GF matrix of a full community that includes all possible community members was used as the nominal input to the trained dCGF-IS. To perturb the nominal GF matrix, each GF was turned on (off, respectively) if it was off (on, respectively) in the nominal GF matrix. The perturbed community function was then predicted using the trained dCGF-IS and the perturbed GF matrix. The difference between nominal community function and perturbed community function was used to evaluate sensitivity of a GF in a species. Finally, the sensitivity of a GF is computed by averaging across all species and all random seeds (not drawn in figure). See **Methods** for more details. (B) Sensitivities of GFs with the top 5 % magnitudes computed from dCGF-IS for the butyrate dataset. Species where the GFs are present are labeled on top of each GF. Species that were found to positively (negatively, respectively) impact community butyrate production on average were labeled red (blue, respectively). The impact of a species on community butyrate production is the difference between the average butyrate in communities with and without this species. GFs that are known to positively (negatively) impact microbial butyrate production are shown in green (brown, respectively).

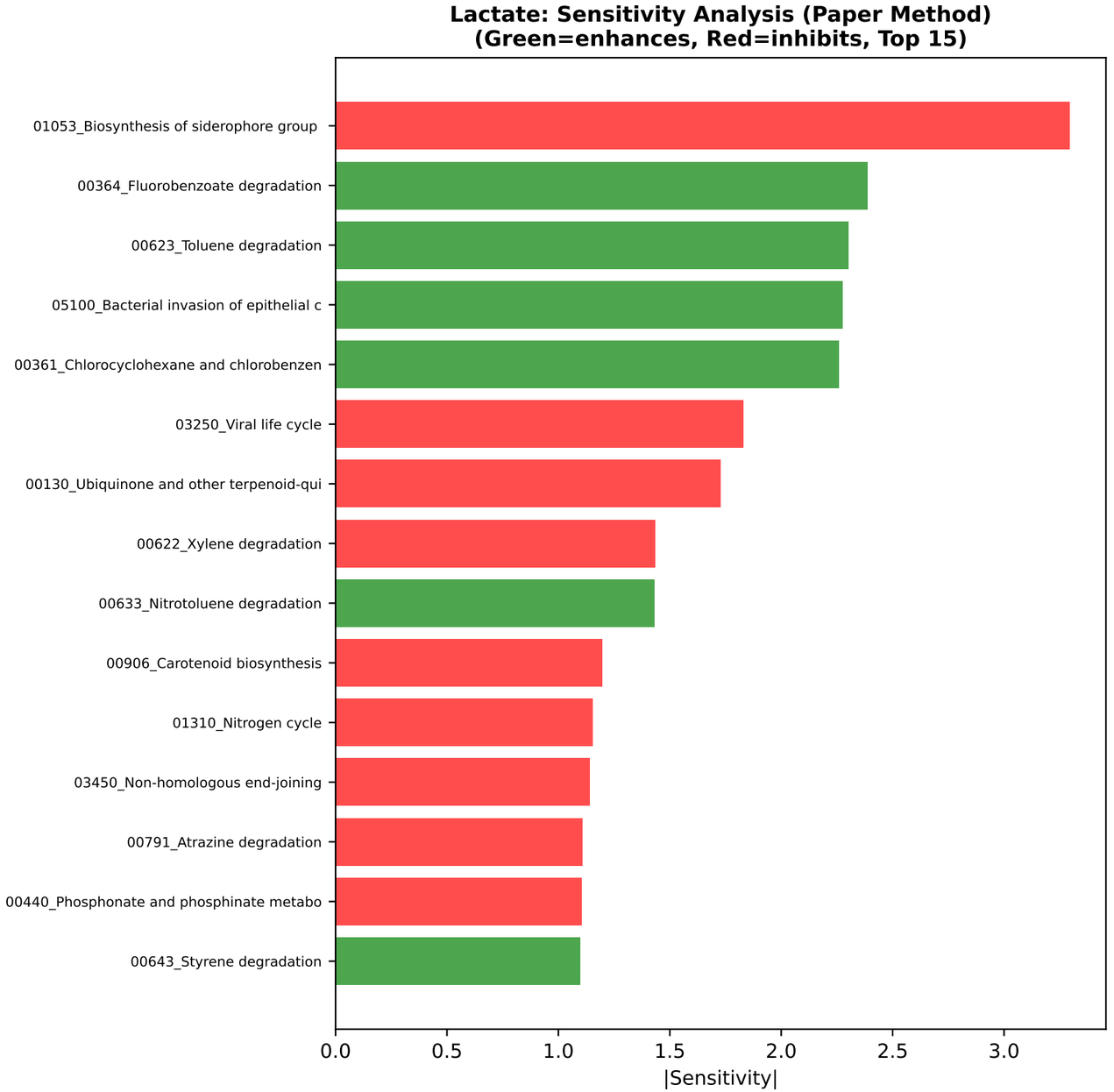

Acetate: Permutation Importance
(Top 15 pathways)
00542_O-Antigen repeat unit biosynthesis
00785_Lipoic acid metabolism
00566_Sulfoquinovose metabolism
00310_Lysine degradation
00130_Ubiquinone and other terpenoid-qui
02030_Bacterial chemotaxis
00121_Secondary bile acid biosynthesis
00311_Penicillin and cephalosporin biosy
04141_Protein processing in endoplasmic
00998_Biosynthesis of various antibiotic
00626_Naphthalene degradation
00380_Tryptophan metabolism
00627_Aminobenzoate degradation
01055_Biosynthesis of vancomycin group a
04382_Cornified envelope formation
0.0000
0.0001
0.0002
0.0003
0.0004
0.0005
0.0006
Importance (drop in r)

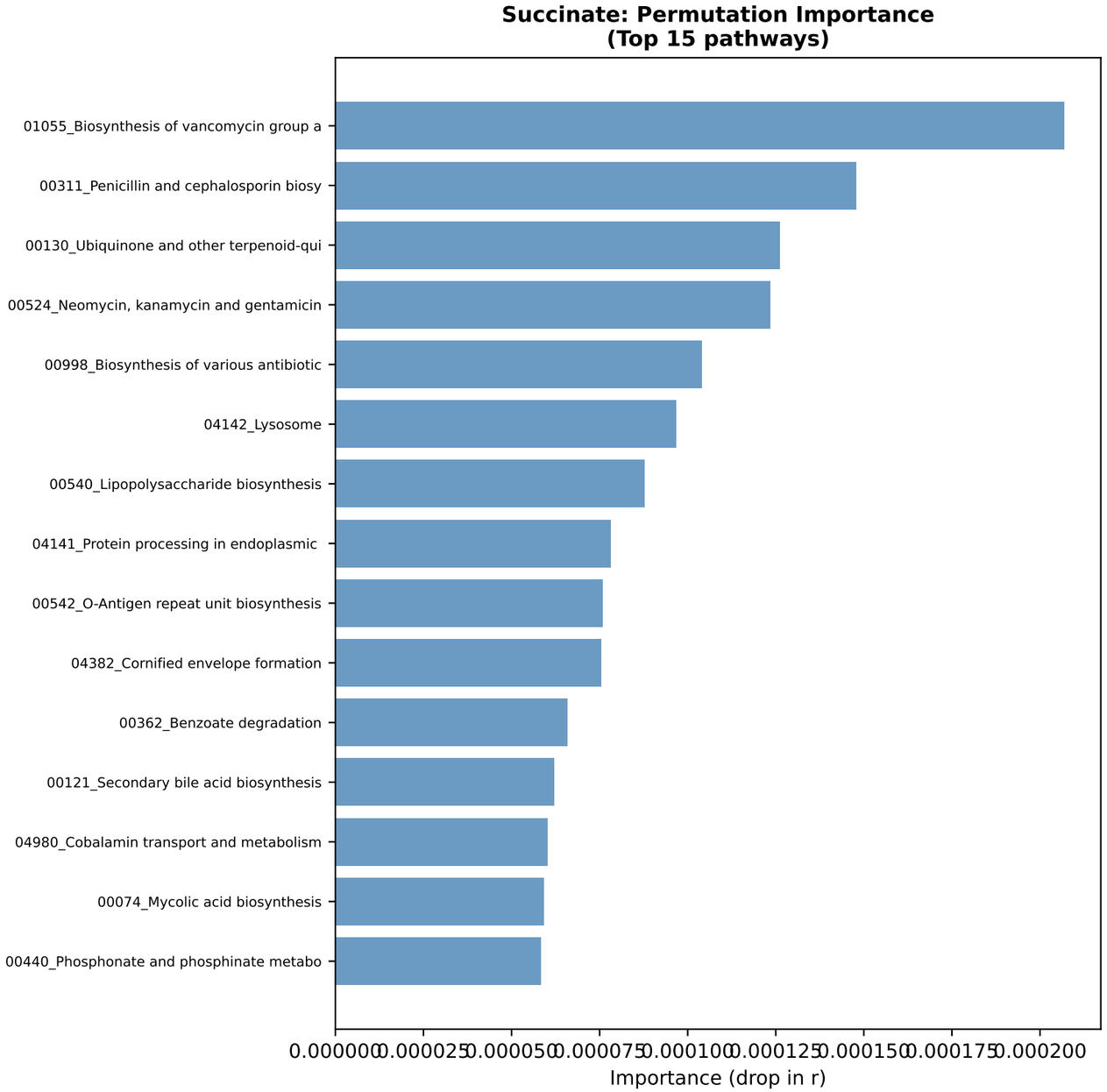

Acetate: Sensitivity Analysis (Paper Method)
(Green=enhances, Red=inhibits, Top 15)
01053_Biosynthesis of siderophore group
00364_Fluorobenzoate degradation
00623_Toluene degradation
05100_Bacterial invasion of epithelial c
00361_Chlorocyclohexane and chlorobenzen
00130_Ubiquinone and other terpenoid-qui
00791_Atrazine degradation
03250_Viral life cycle
03450_Non-homologous end-joining
00906_Carotenoid biosynthesis
00121_Secondary bile acid biosynthesis
00440_Phosphonate and phosphinate metabo
03008_Ribosome biogenesis in eukaryotes
01310_Nitrogen cycle
00999_Biosynthesis of various plant seco
0.0
0.5
1.0
1.5
2.0
2.5
|Sensitivity|

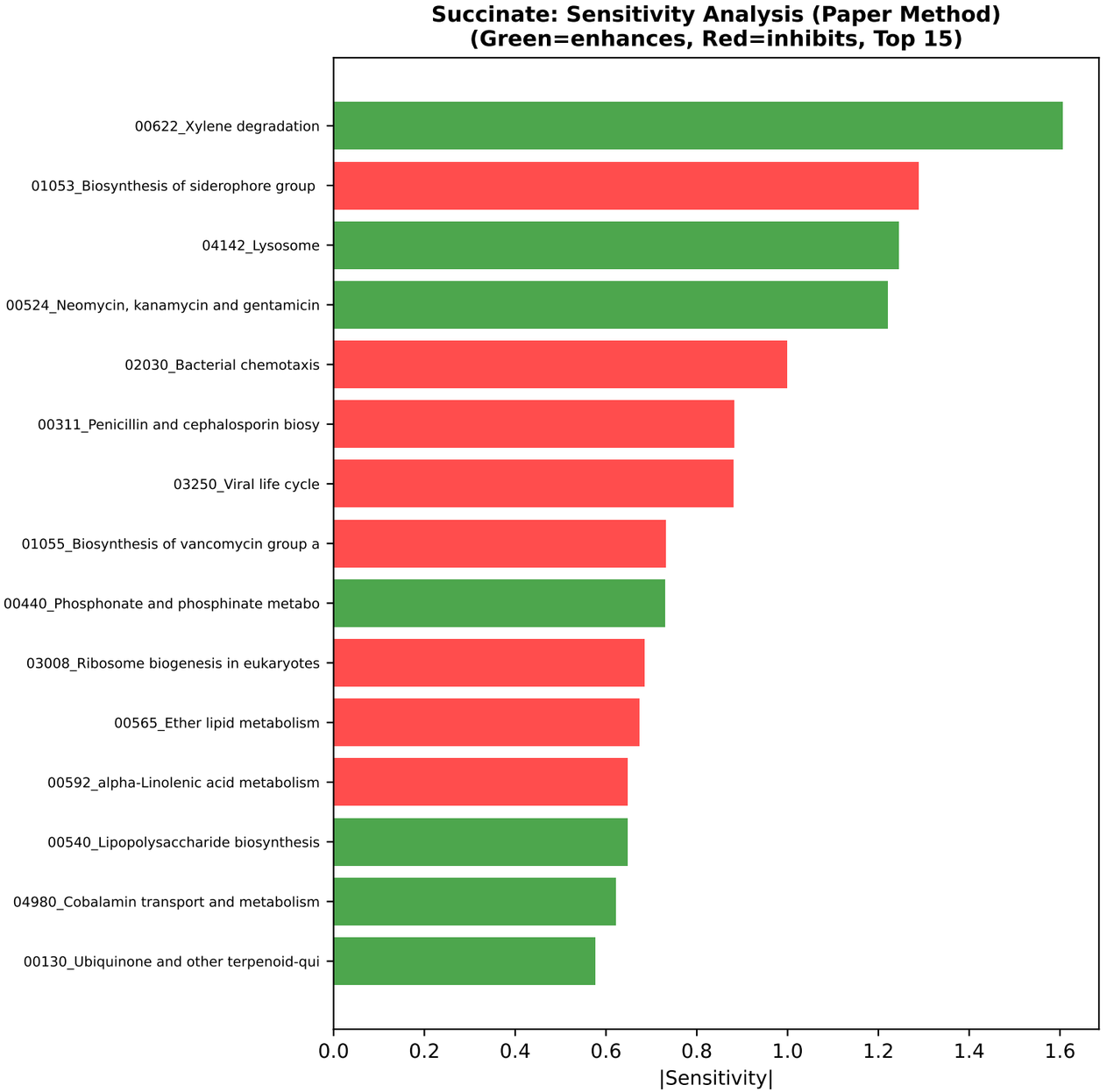

Butyrate: Permutation Importance
(Top 15 pathways)
00626_Naphthalene degradation
01220_Degradation of aromatic compounds
00627_Aminobenzoate degradation
00791_Atrazine degradation
00310_Lysine degradation
02030_Bacterial chemotaxis
00999_Biosynthesis of various plant seco
00380_Tryptophan metabolism
00998_Biosynthesis of various antibiotic
00562_Inositol phosphate metabolism
01320_Sulfur cycle
00130_Ubiquinone and other terpenoid-qui
00785_Lipoic acid metabolism
04142_Lysosome
00542_O-Antigen repeat unit biosynthesis
0.00000 0.00002 0.00004 0.00006 0.00008 0.00010 0.00012 0.00014 0.00016
Importance (drop in r)


Butyrate: Sensitivity Analysis (Paper Method)
(Green=enhances, Red=inhibits, Top 15)
01053_Biosynthesis of siderophore group
03250_Viral life cycle
00361_Chlorocyclohexane and chlorobenzen
00623_Toluene degradation
05100_Bacterial invasion of epithelial c
00364_Fluorobenzoate degradation
00130_Ubiquinone and other terpenoid-qui
00633_Nitrotoluene degradation
00311_Penicillin and cephalosporin biosy
00643_Styrene degradation
00460_Cyanoamino acid metabolism
00791_Atrazine degradation
00566_Sulfoquinovose metabolism
00542_O-Antigen repeat unit biosynthesis
00121_Secondary bile acid biosynthesis
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
|Sensitivity|

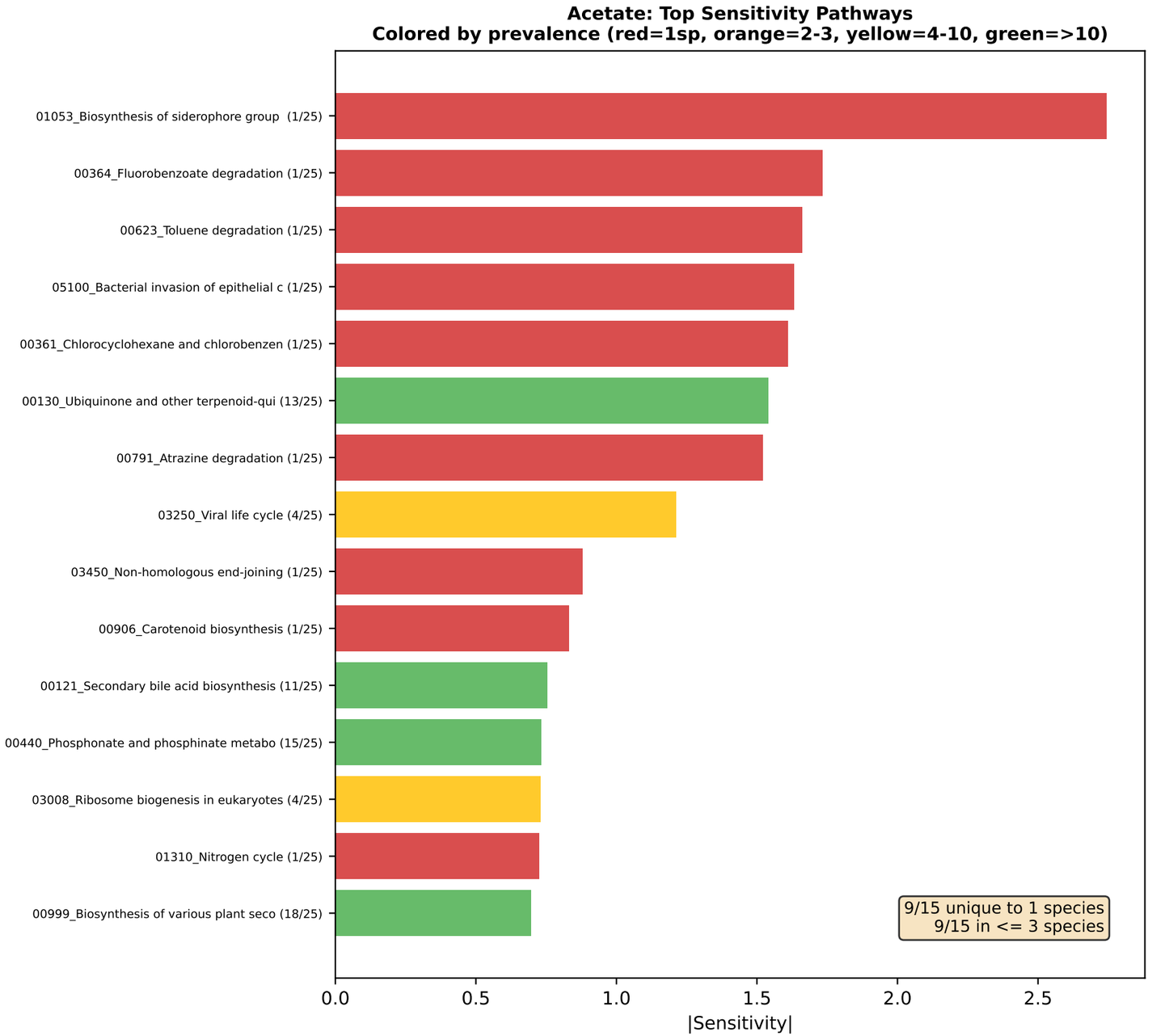

Lactate: Permutation Importance
(Top 15 pathways)
03250_Viral life cycle
00542_O-Antigen repeat unit biosynthesis
01220_Degradation of aromatic compounds
00633_Nitrotoluene degradation
03008_Ribosome biogenesis in eukaryotes
00380_Tryptophan metabolism
00626_Naphthalene degradation
00121_Secondary bile acid biosynthesis
00074_Mycolic acid biosynthesis
00130_Ubiquinone and other terpenoid-qui
00627_Aminobenzoate degradation
00643_Styrene degradation
01320_Sulfur cycle
01054_Nonribosomal peptide structures
00361_Chlorocyclohexane and chlorobenzen
0.00000
0.00005
0.00010
0.00015
0.00020
0.00025
0.00030
Importance (drop in r)

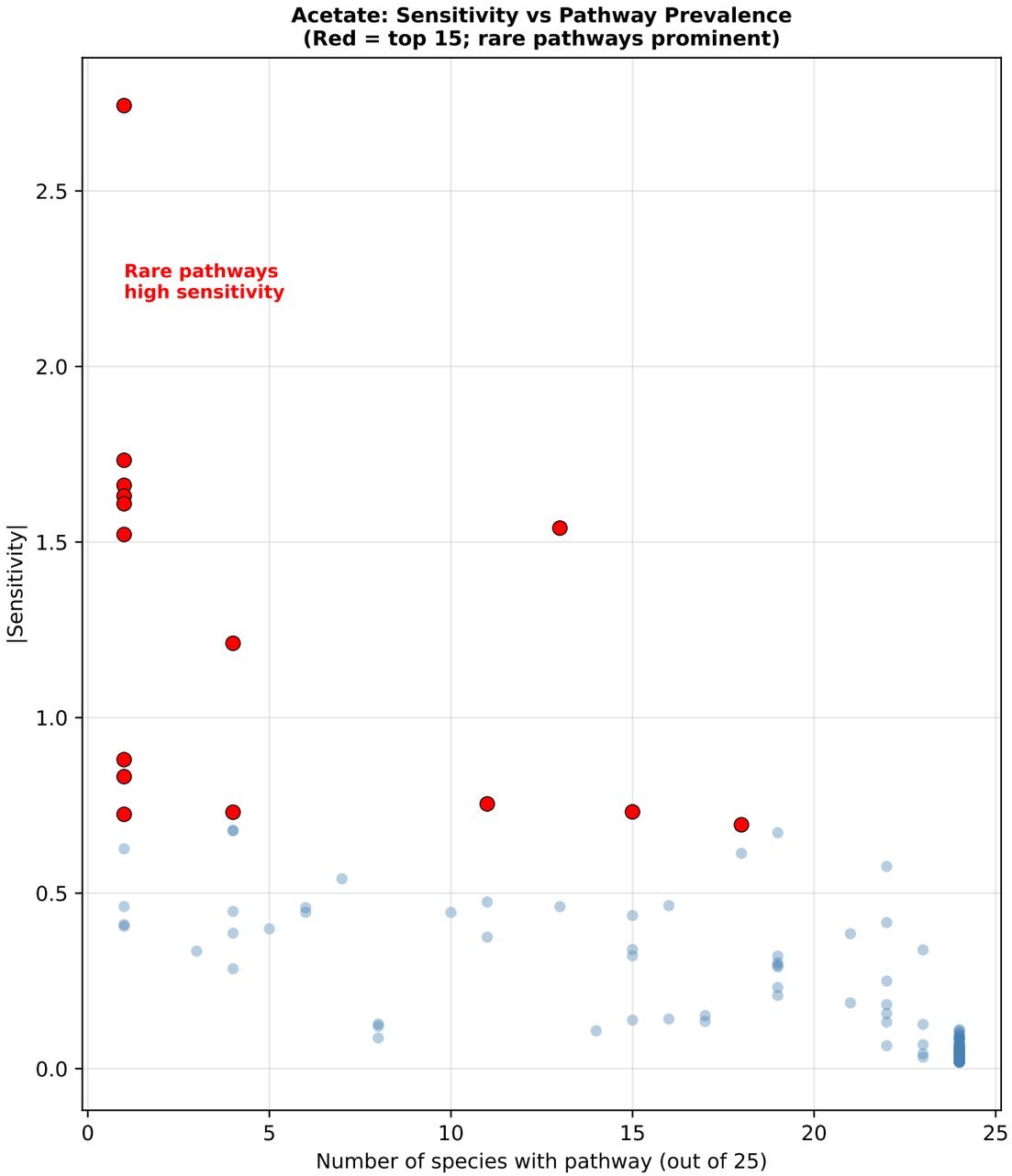

Lactate: Sensitivity Analysis (Paper Method)
(Green=enhances, Red=inhibits, Top 15)
01053_Biosynthesis of siderophore group
00364_Fluorobenzoate degradation
00623_Toluene degradation
05100_Bacterial invasion of epithelial c
00361_Chlorocyclohexane and chlorobenzen
03250_Viral life cycle
00130_Ubiquinone and other terpenoid-qui
00622_Xylene degradation
00633_Nitrotoluene degradation
00906_Carotenoid biosynthesis
01310_Nitrogen cycle
03450_Non-homologous end-joining
00791_Atrazine degradation
00440_Phosphonate and phosphinate metabo
00643_Styrene degradation
0.0
0.5
1.0
1.5
2.0
2.5
3.0
|Sensitivity|


Succinate: Permutation Importance
(Top 15 pathways)
01055_Biosynthesis of vancomycin group a
00311_Penicillin and cephalosporin biosy
00130_Ubiquinone and other terpenoid-qui
00524_Neomycin, kanamycin and gentamicin
00998_Biosynthesis of various antibiotic
04142_Lysosome
00540_Lipopolysaccharide biosynthesis
04141_Protein processing in endoplasmic
00542_O-Antigen repeat unit biosynthesis
04382_Cornified envelope formation
00362_Benzoate degradation
00121_Secondary bile acid biosynthesis
04980_Cobalamin transport and metabolism
00074_Mycolic acid biosynthesis
00440_Phosphonate and phosphinate metabo
0.000000
0.000025
0.000050
0.000075
0.000100
0.000125
0.000150
0.000175
0.000200
Importance (drop in r)

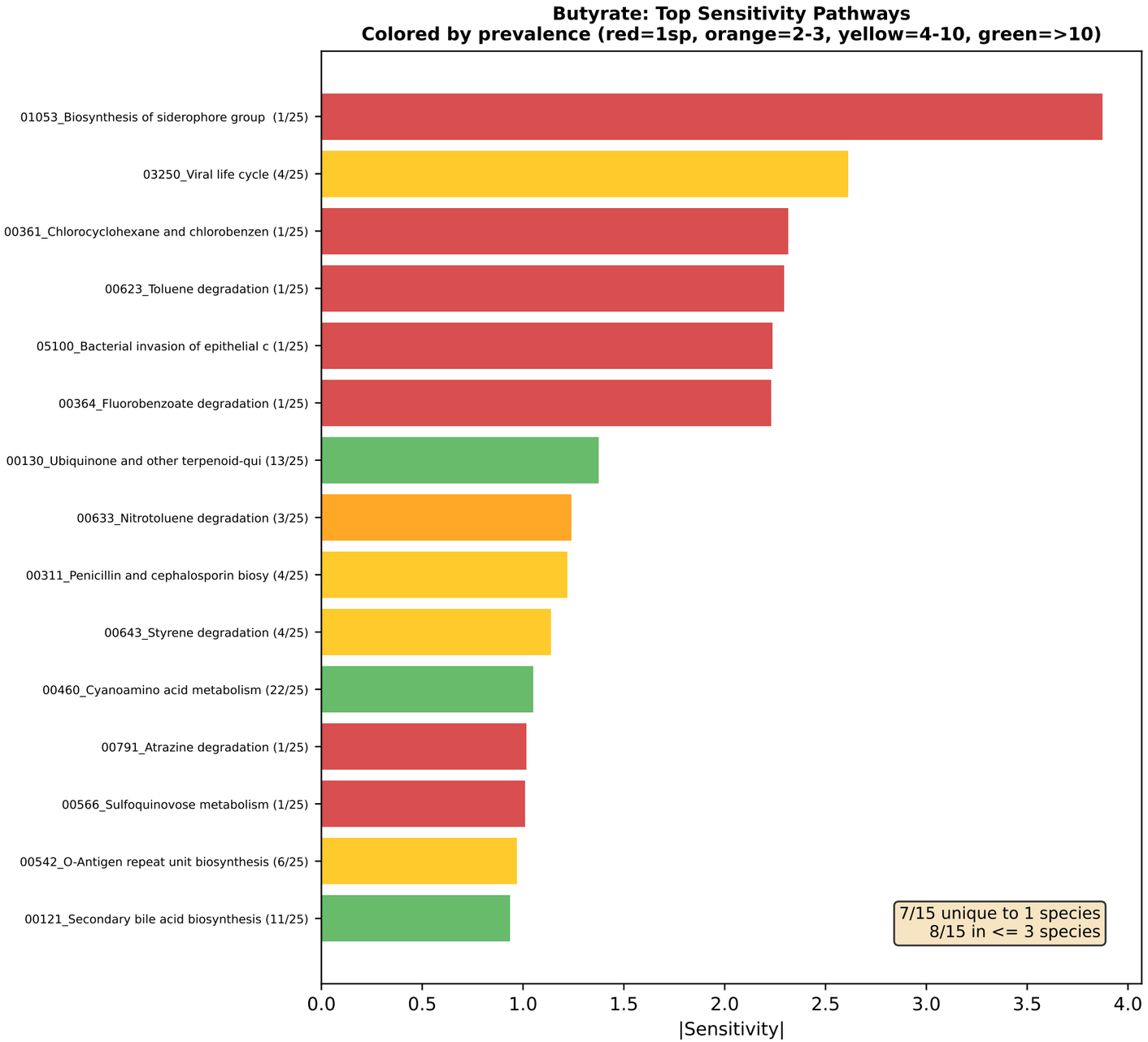

Succinate: Sensitivity Analysis (Paper Method)
(Green=enhances, Red=inhibits, Top 15)
00622_Xylene degradation
01053_Biosynthesis of siderophore group
04142_Lysosome
00524_Neomycin, kanamycin and gentamicin
02030_Bacterial chemotaxis
00311_Penicillin and cephalosporin biosy
03250_Viral life cycle
01055_Biosynthesis of vancomycin group a
00440_Phosphonate and phosphinate metabo
03008_Ribosome biogenesis in eukaryotes
00565_Ether lipid metabolism
00592_alpha-Linolenic acid metabolism
00540_Lipopolysaccharide biosynthesis
04980_Cobalamin transport and metabolism
00130_Ubiquinone and other terpenoid-qui
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
|Sensitivity|

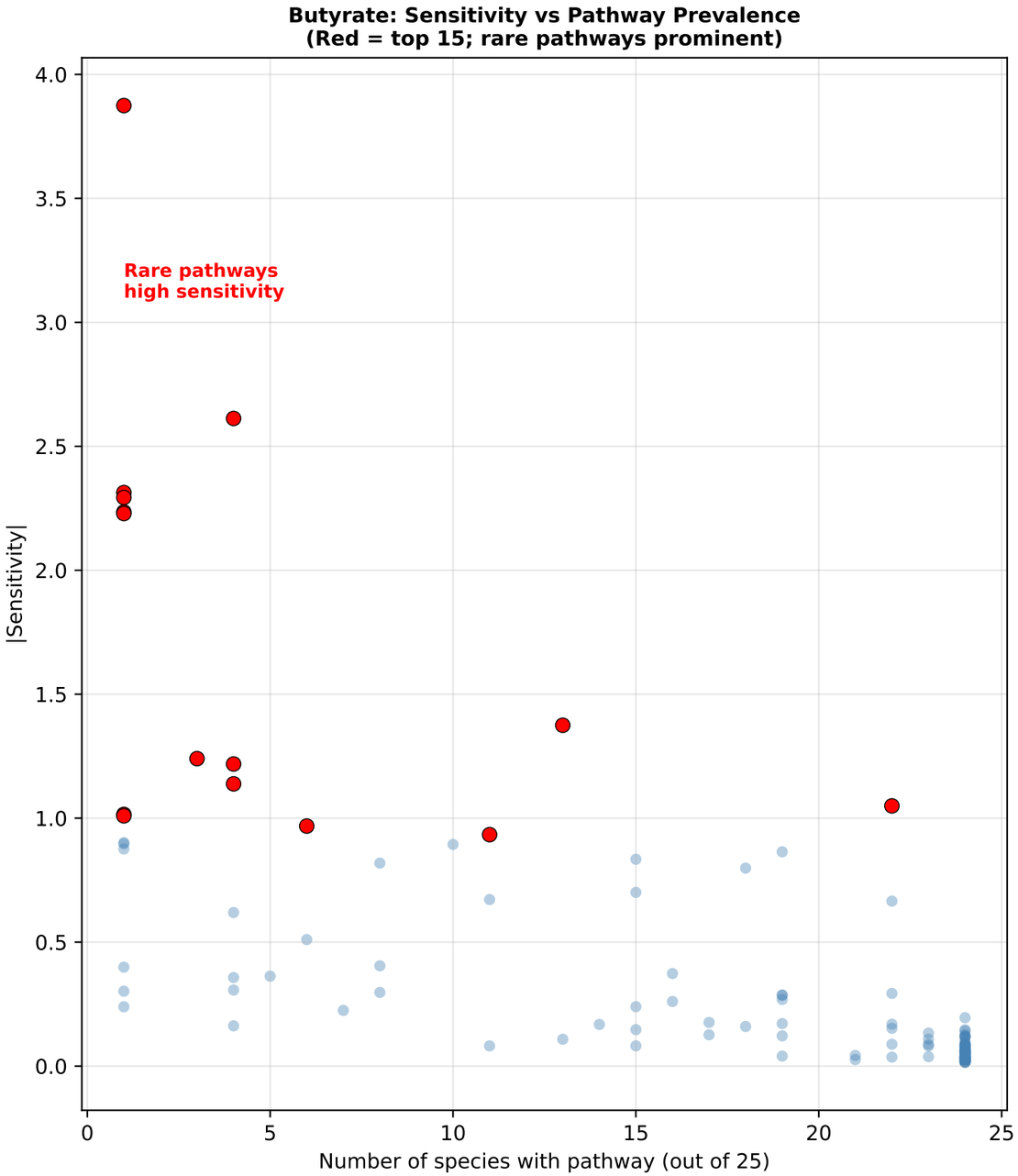
Sensitivity Analysis: Relationship Between Pathway Prevalence and Importance



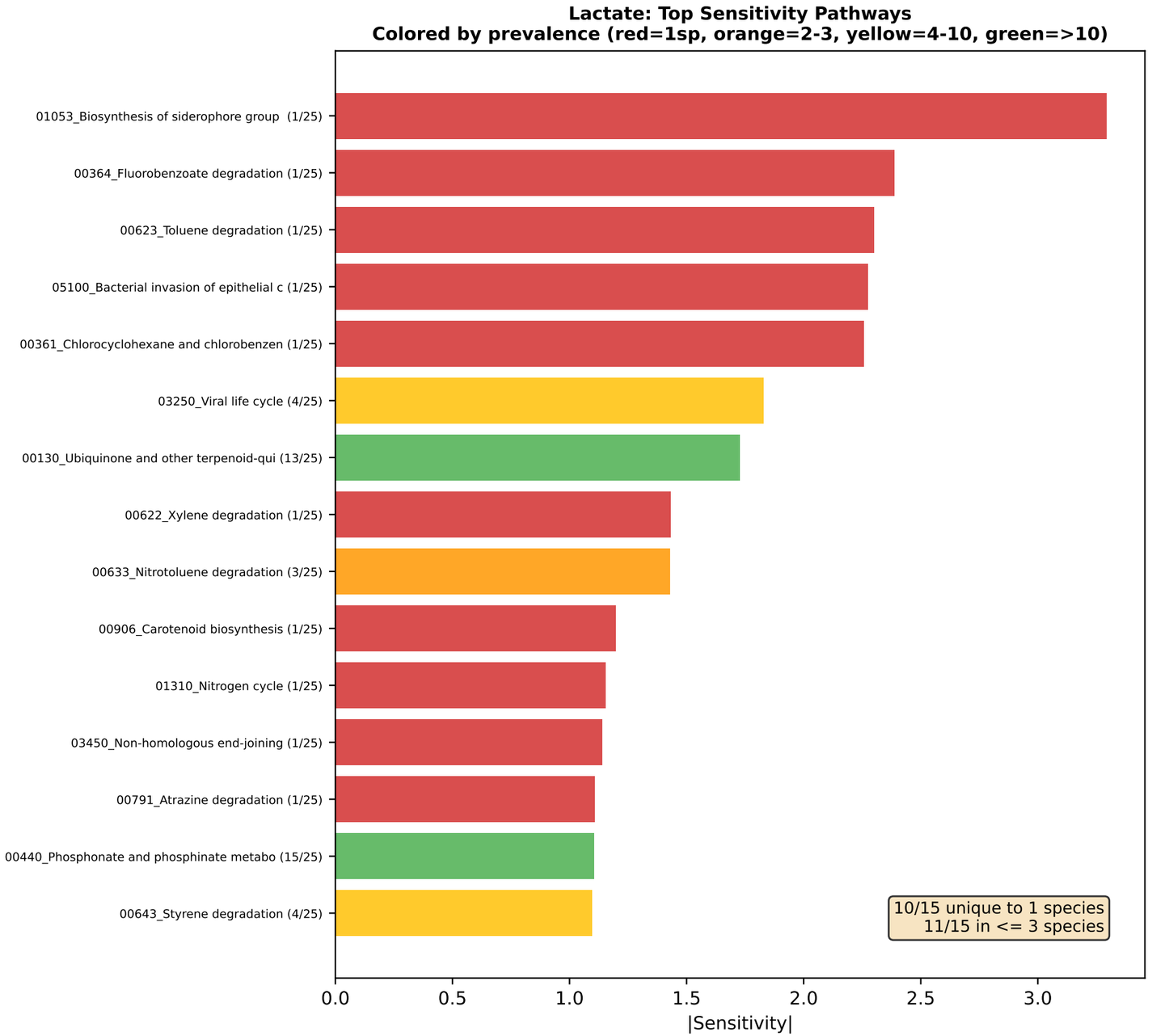


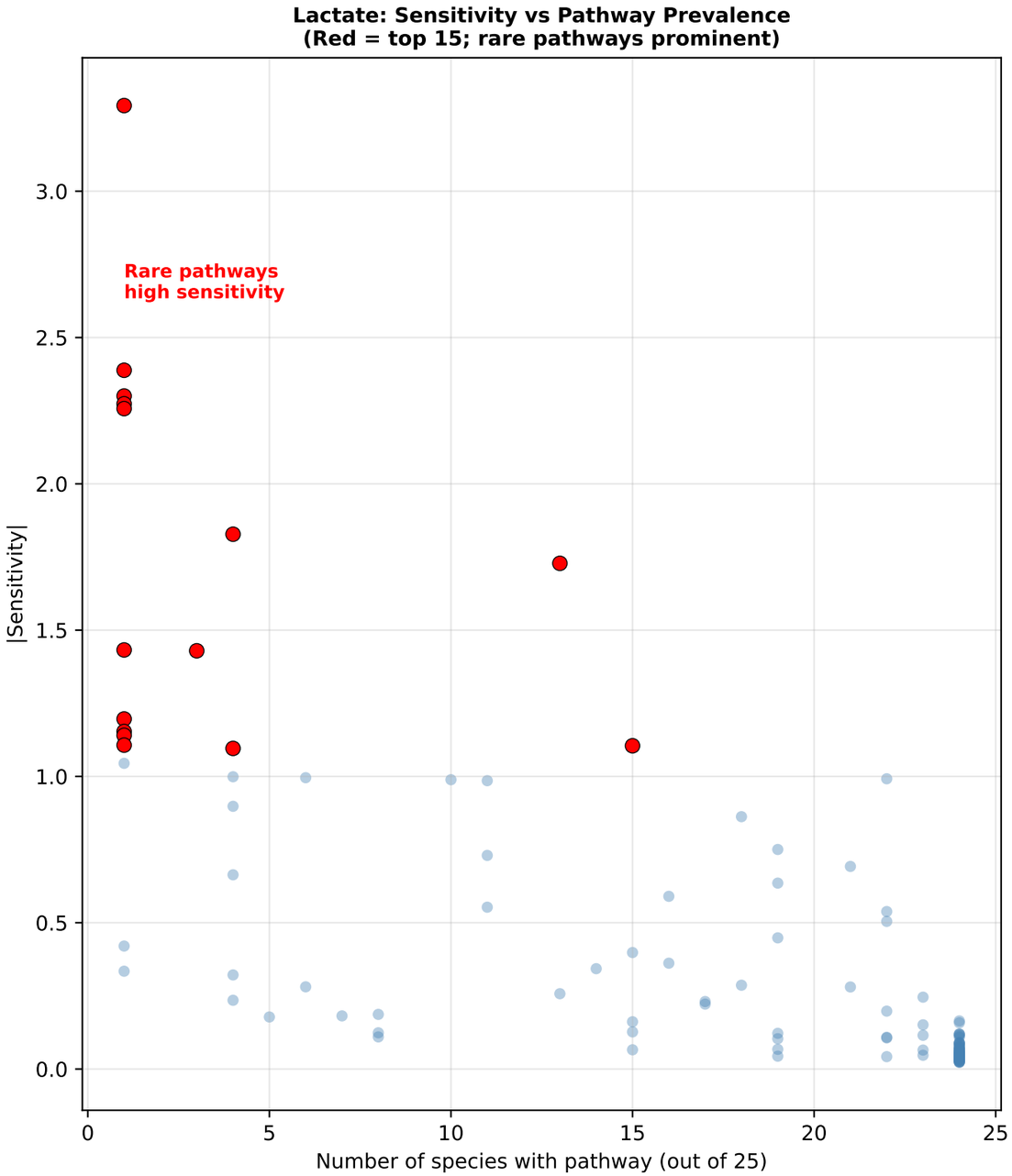
Sensitivity Analysis: Relationship Between Pathway Prevalence and Importance



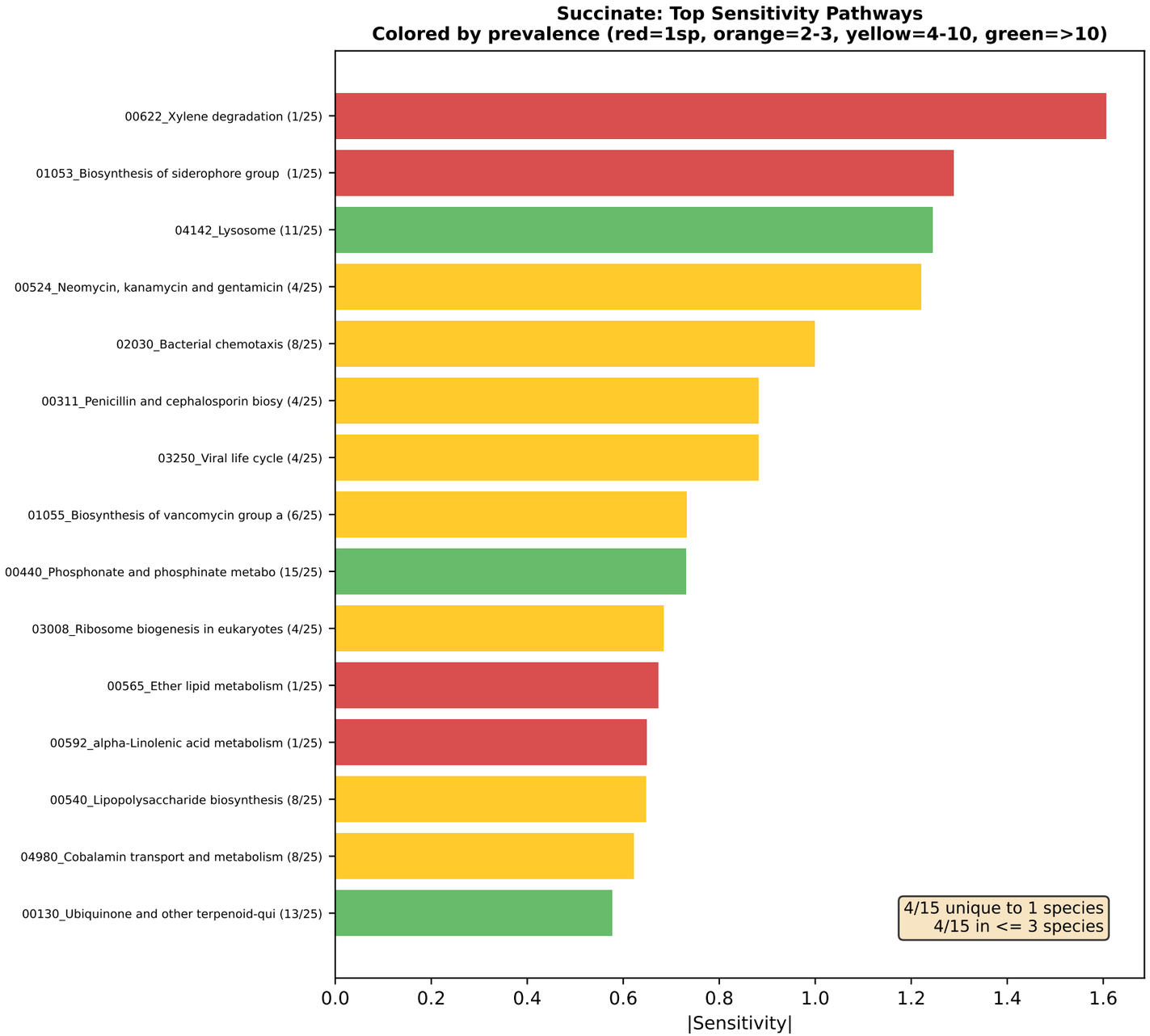


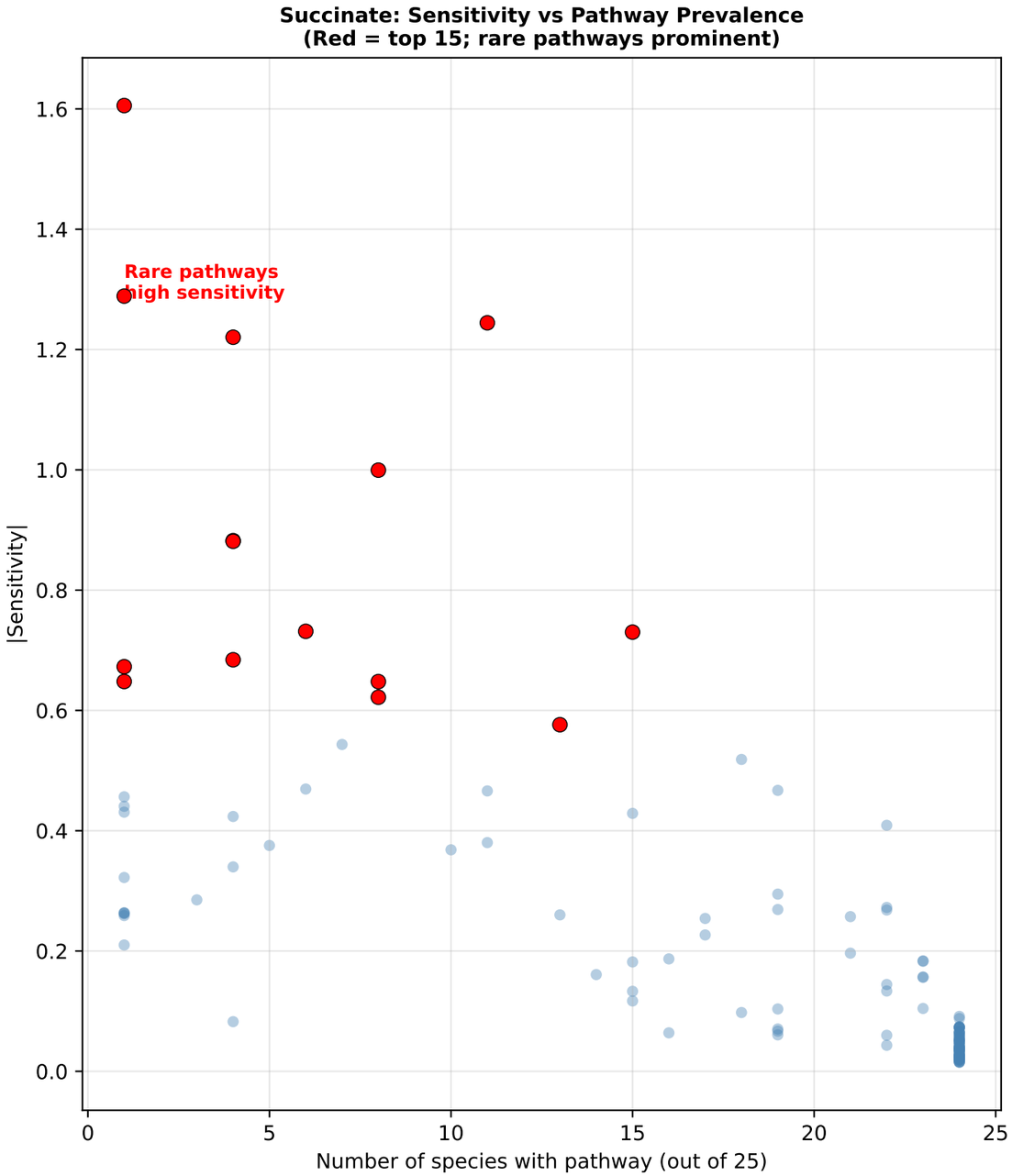
Sensitivity Analysis: Relationship Between Pathway Prevalence and Importance



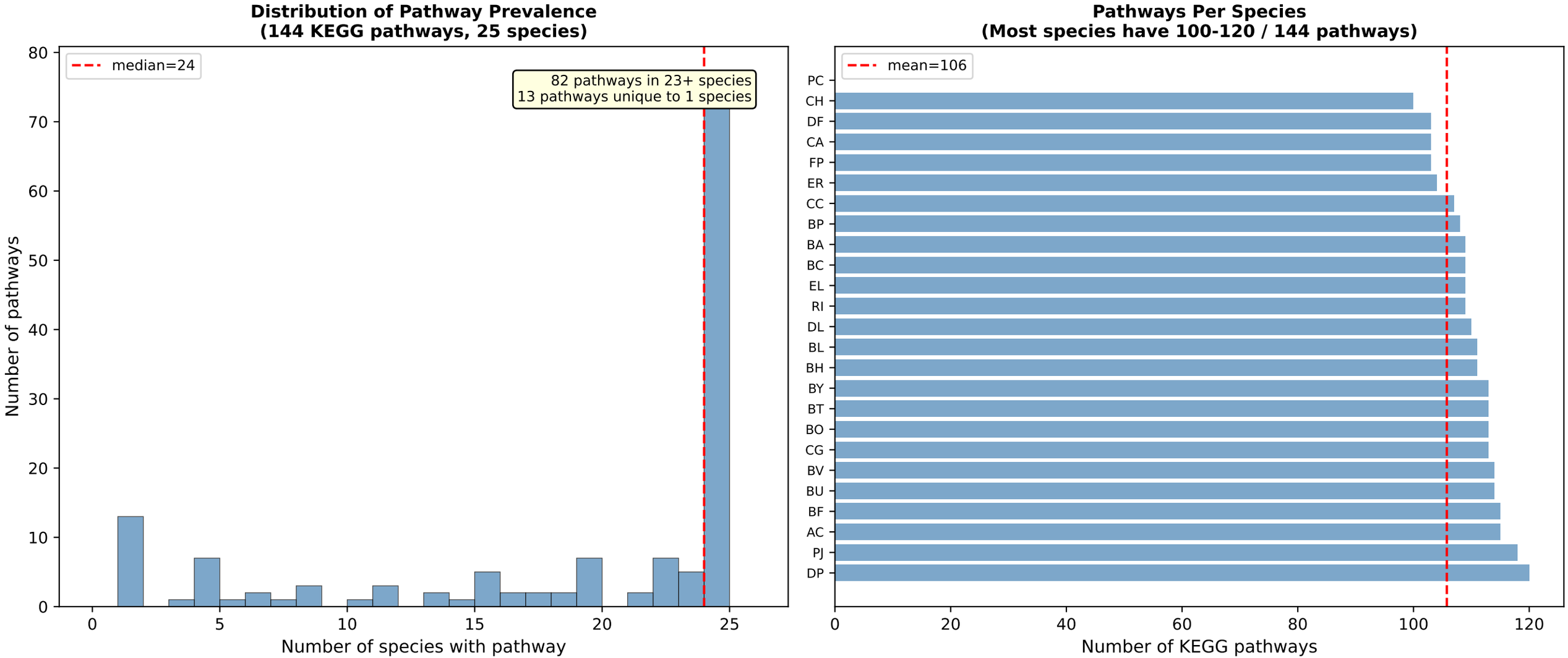


Sensitivity Analysis: Relationship Between Pathway Prevalence and Importance





Pathway Feature Space: Distribution Across Species
Most pathways are widely shared; a subset is species-specific



Paper Figure 6: PCA of GF Vectors vs Functional Embeddings (BioCyc pathways, colored by butyrate producer status)

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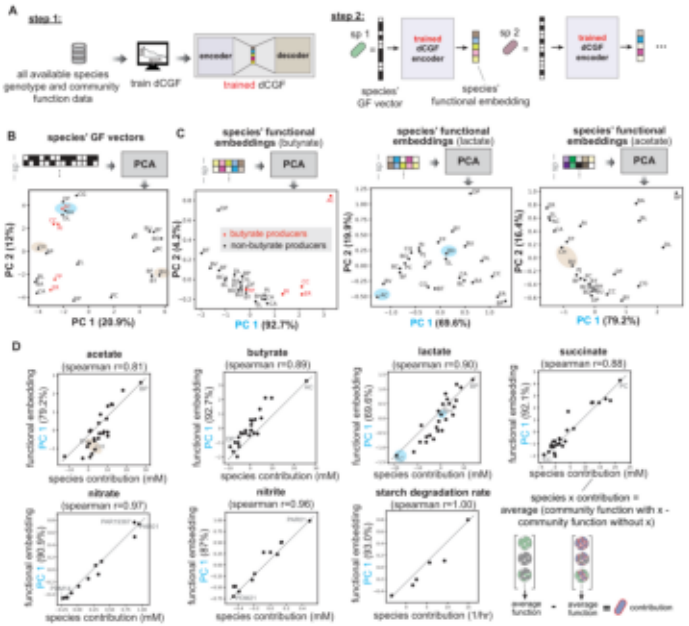
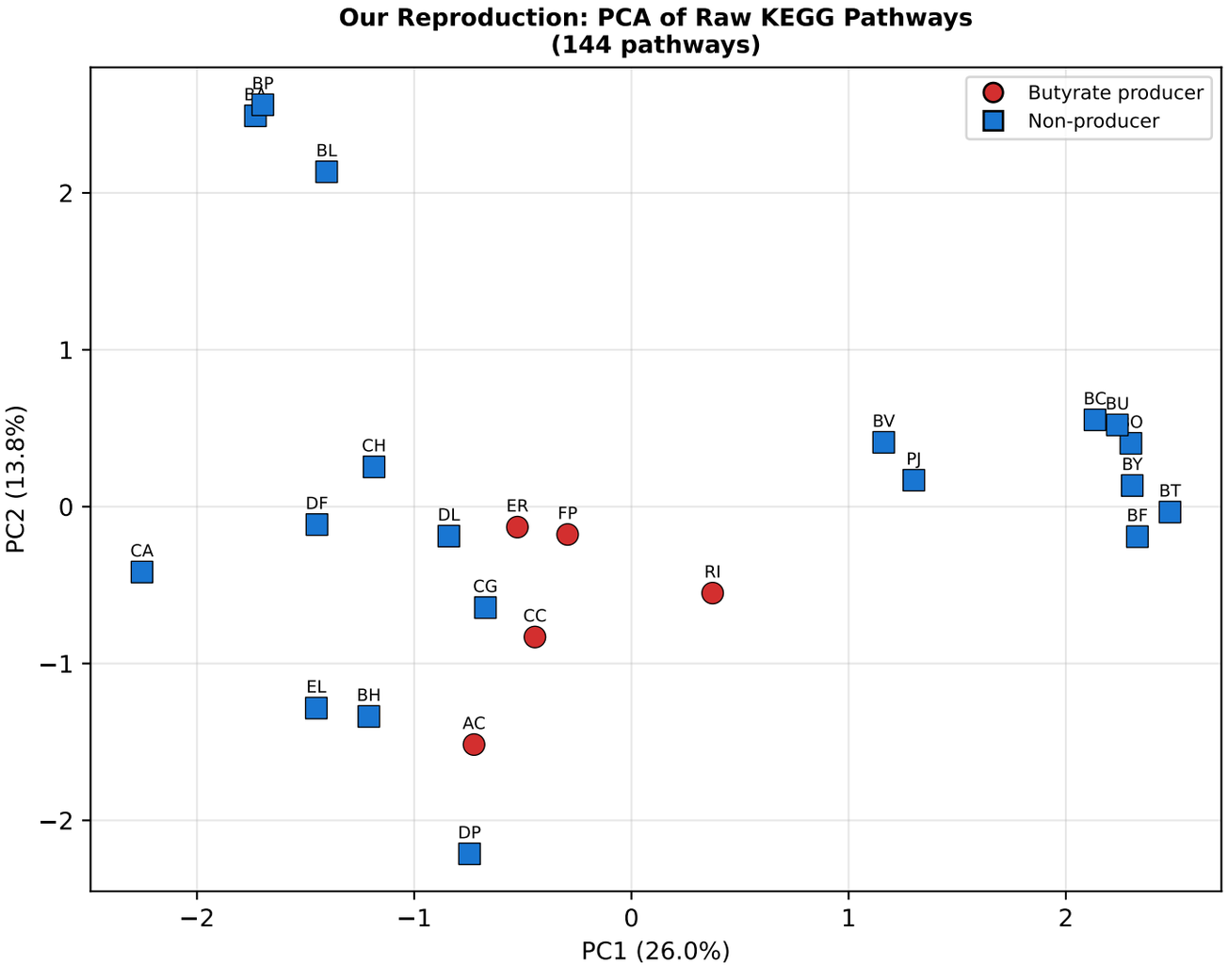


Figure 6: **Species functional embeddings are highly relevant to their functional contributions in the communities.** (A) Species' functional embeddings were derived by first training a dCGF using the entire dataset. The GF vector of each species (with unity abundance) is then fed into the trained dCGF encoder to obtain its functional embedding. (B)-(C) Principal component analysis (PCA) of the species' GF vectors (B) and functional embeddings (C) for butyrate, lacate, and acetate datasets. (D) Correlation between species' scores for PC 1, computed using their functional embeddings, and their contributions to each community functions. The contribution of species x to a community function is computed as the difference between the average measured functions of communities including x and the average measured functions of communities excluding x in the entire dataset.

Figure 6B Reproduction: PCA of Raw Genetic Feature Vectors

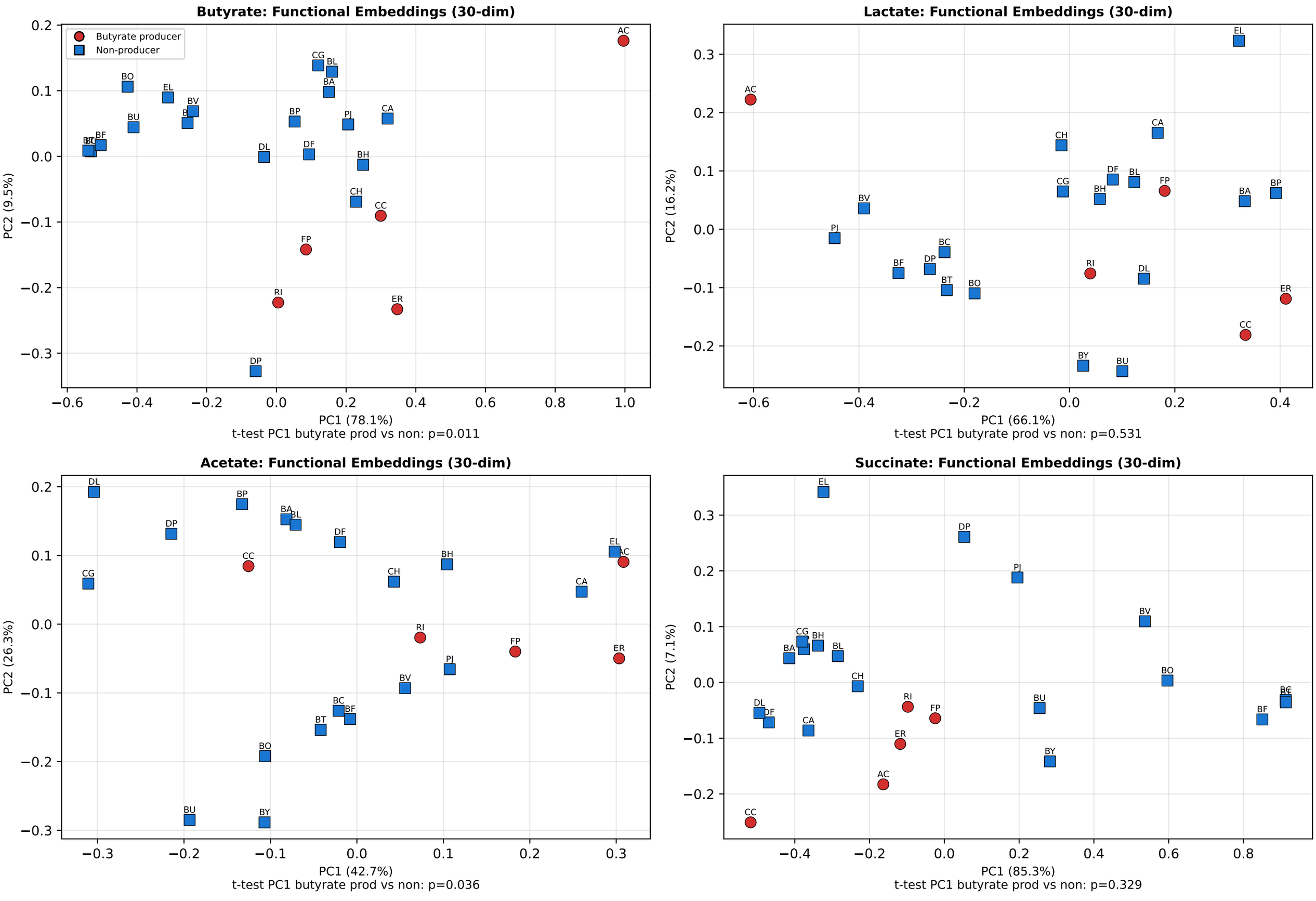


Paper Figure 6B

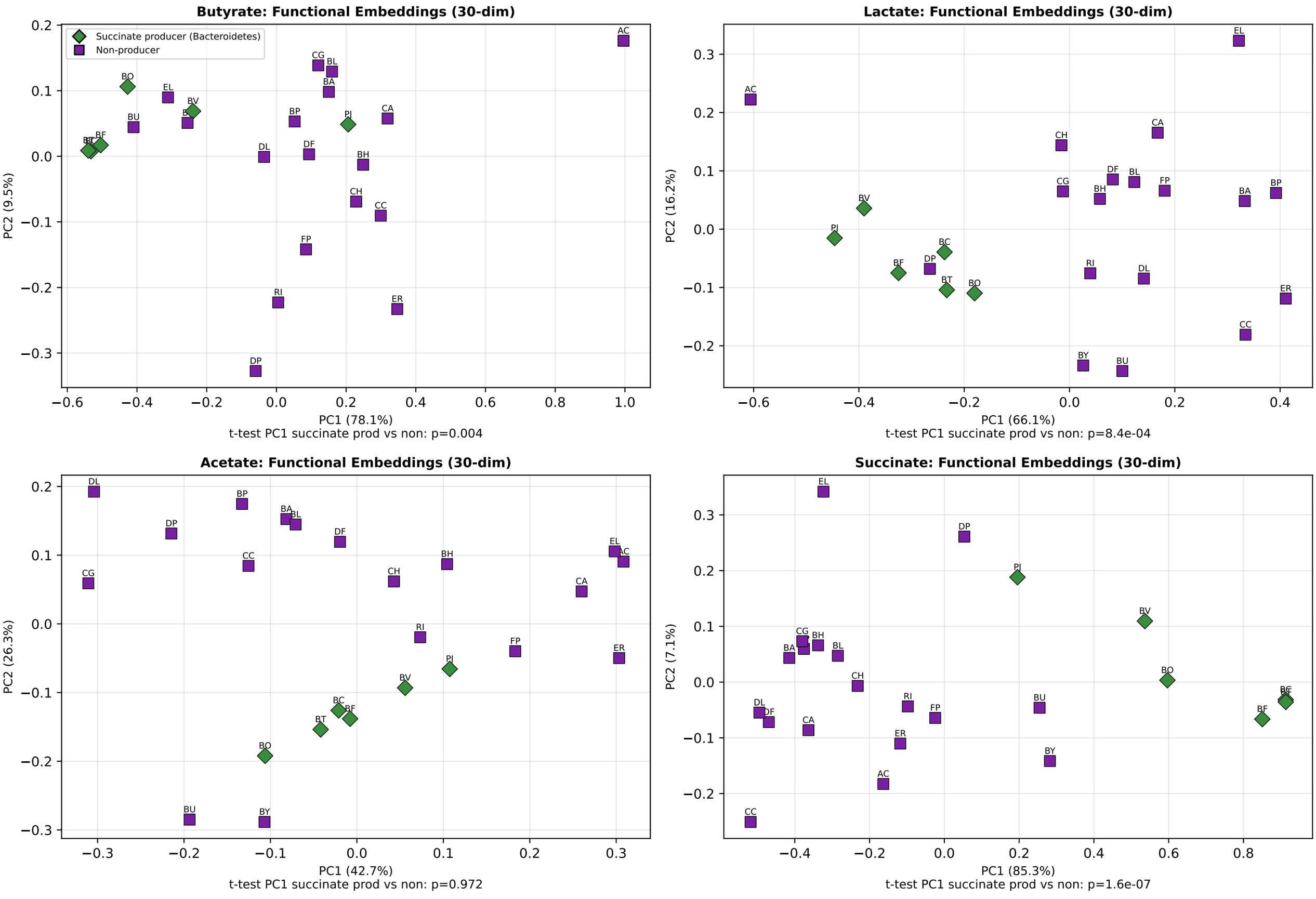
PCA of raw GF vectors (500 BioCyc pathways)
Shows butyrate producers (red) scattered among
non-producers — no clear clustering in
raw pathway space.

Paper reports:
PC1: 20.9% variance
PC2: 12.0% variance

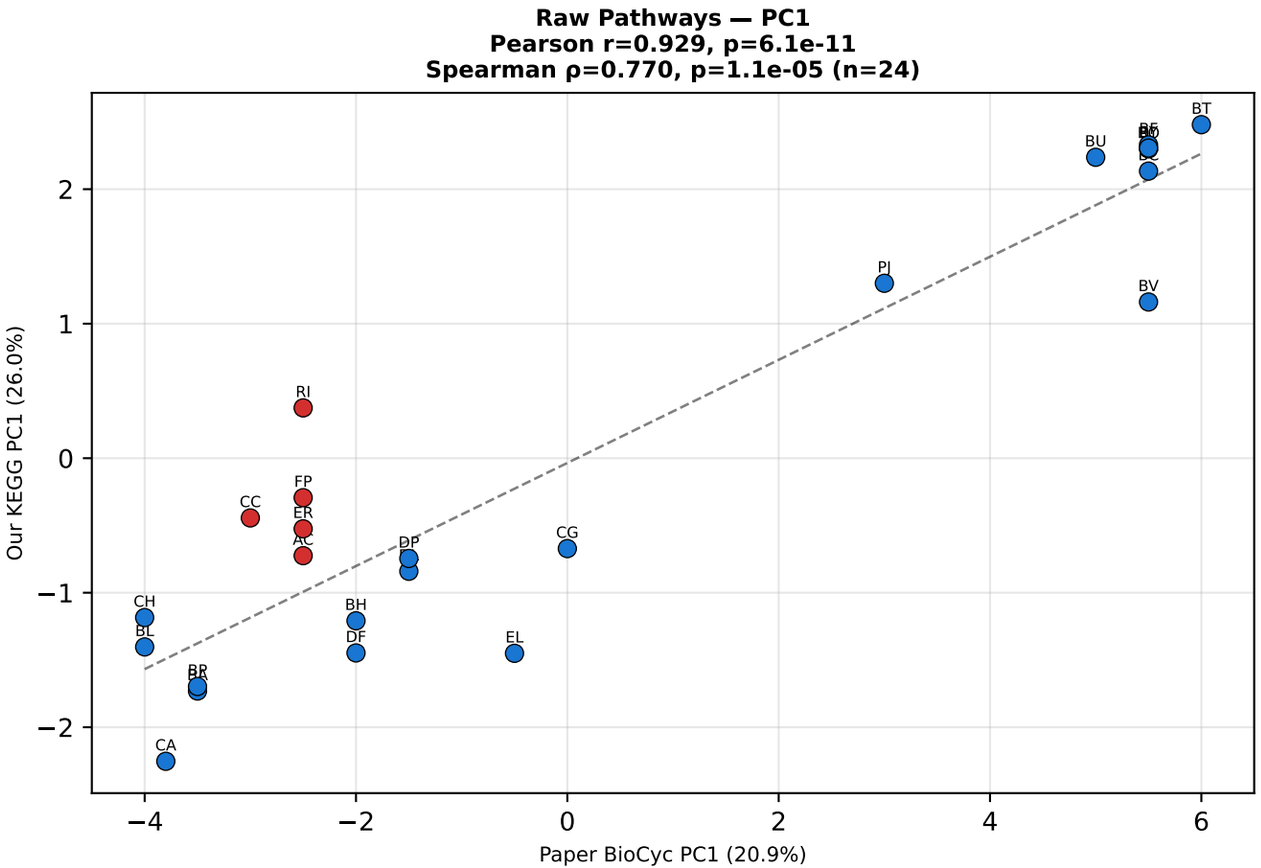
Figure 6C Reproduction: PCA of dCGF-IS Functional Embeddings
(30-dim species embeddings from trained encoder, colored by butyrate producer)

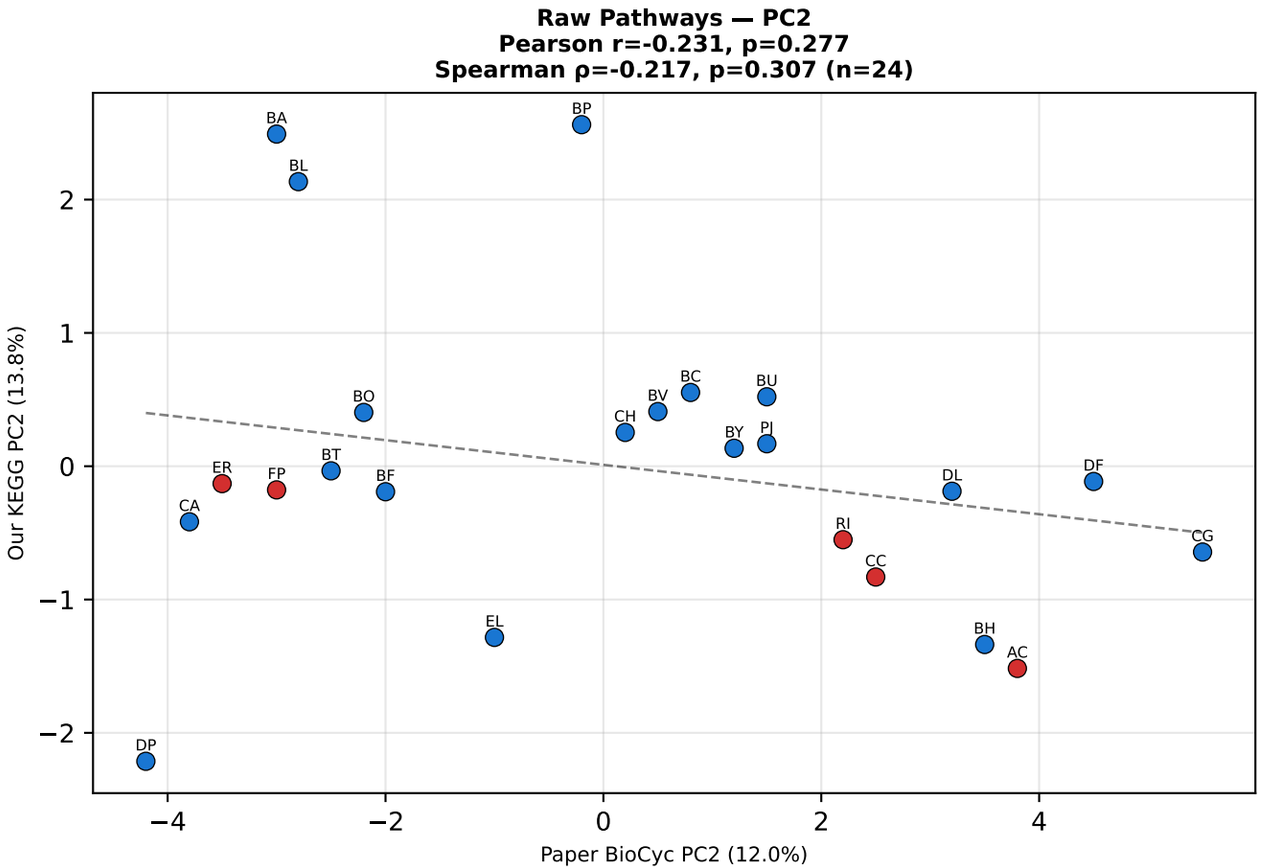


PCA of dCGF-IS Functional Embeddings
(same embeddings as previous page, colored by succinate producer status)

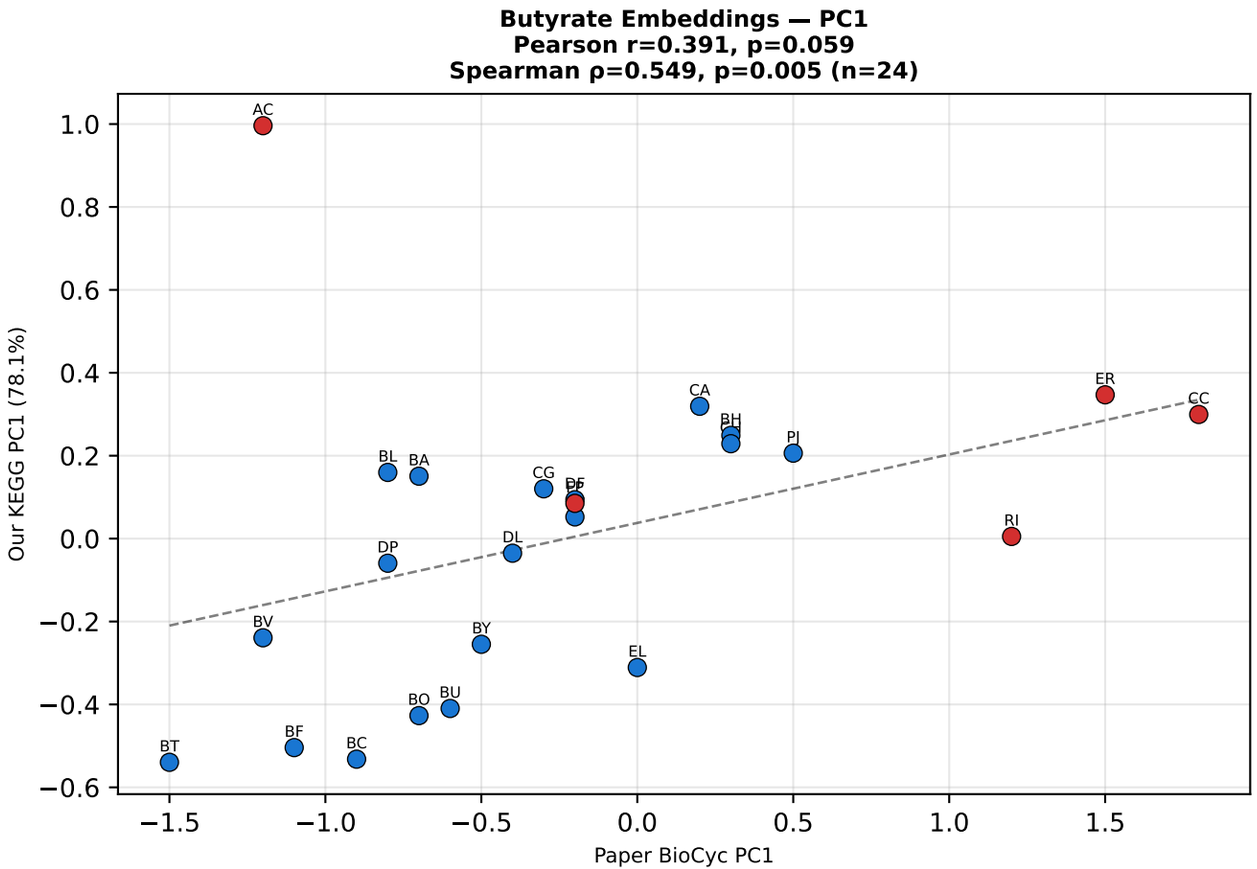


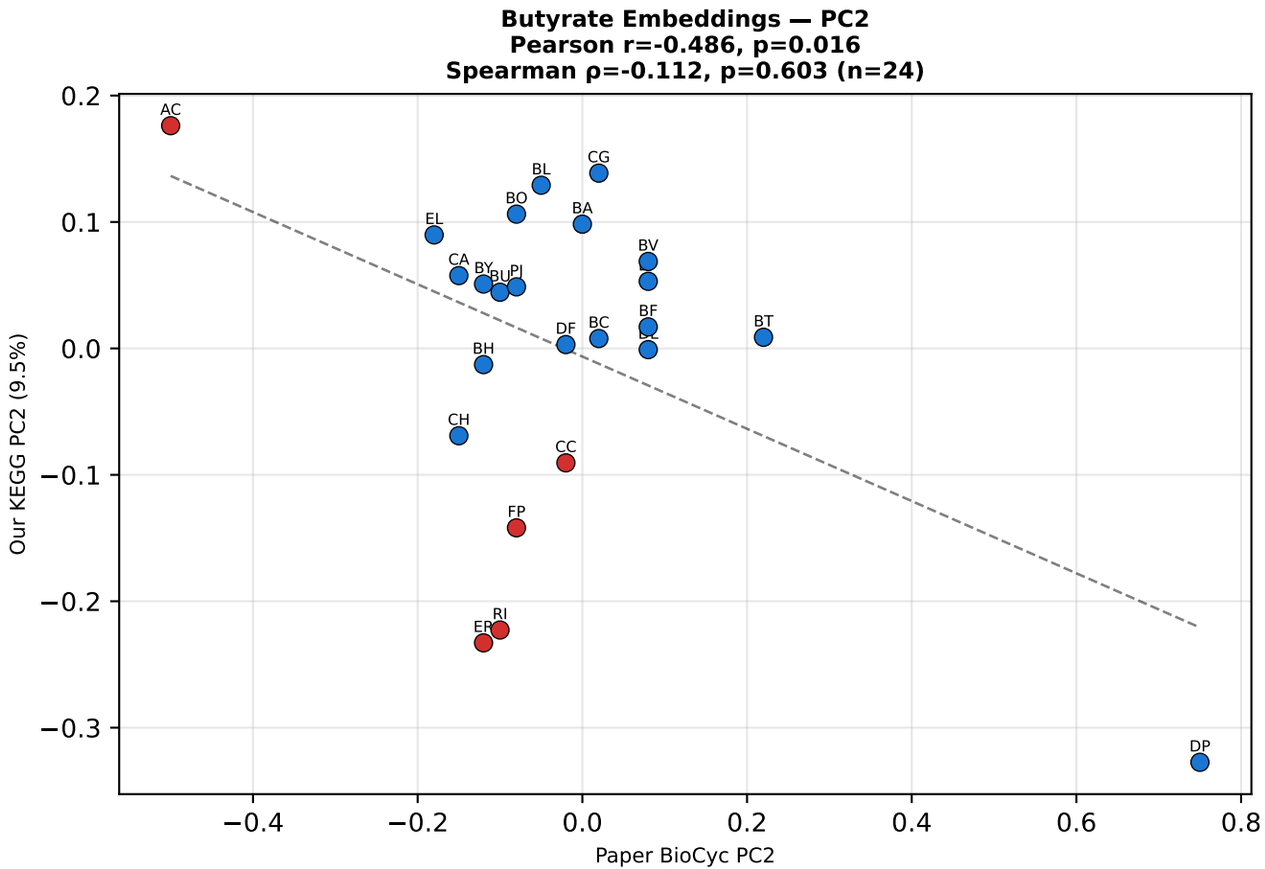
PCA Comparison: Our KEGG Pathways vs Paper BioCyc Pathways
(Raw genetic feature vectors, per-species coordinates)



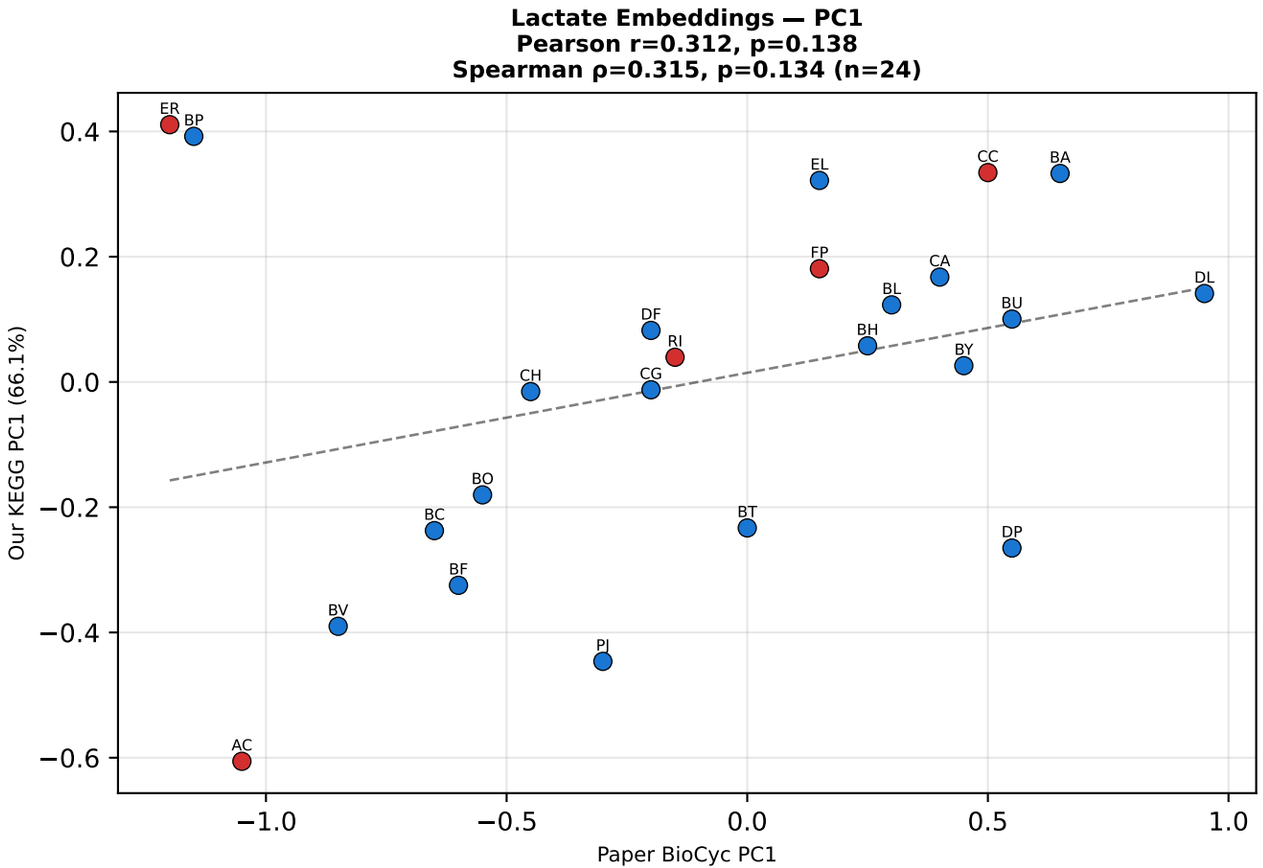


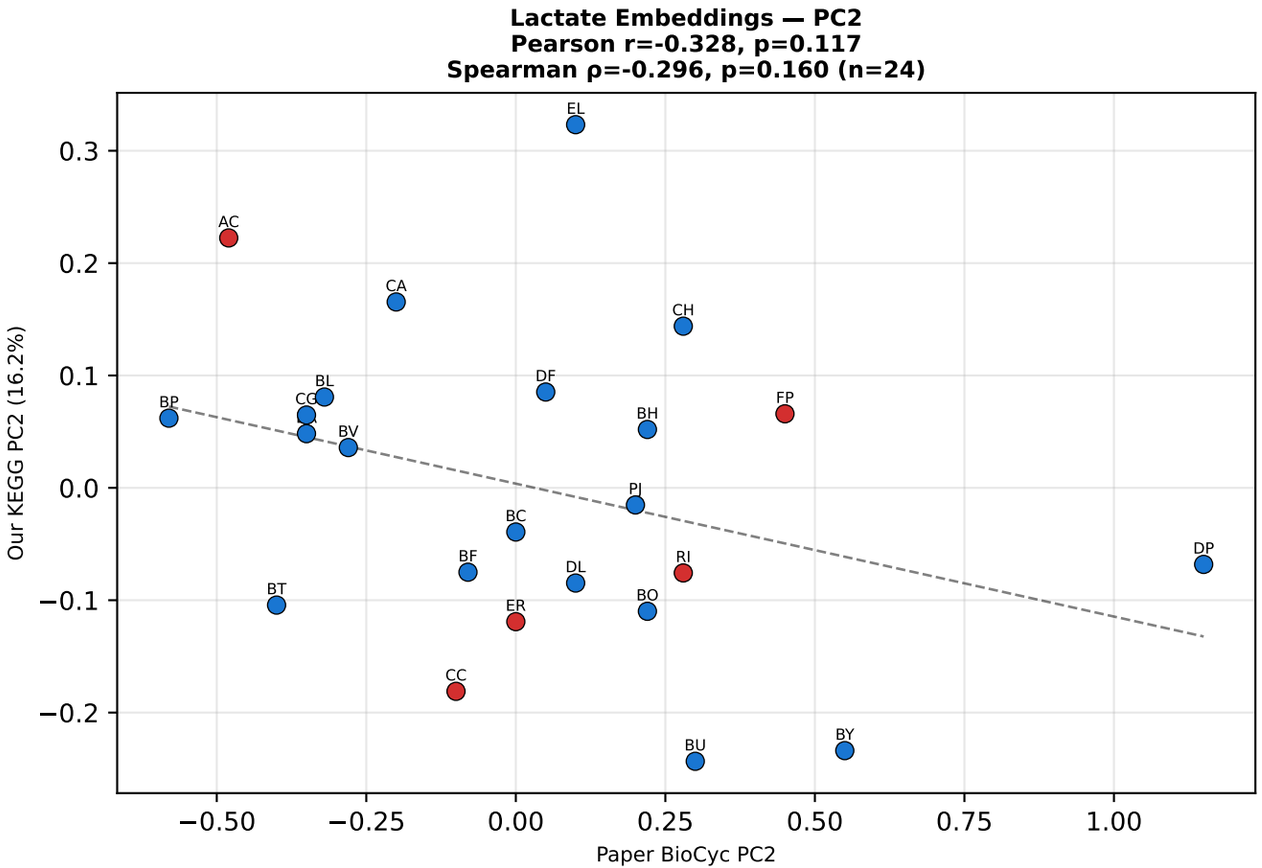
PCA Comparison: Butyrate Functional Embeddings
Our KEGG-trained dCGF-IS vs Paper BioCyc-trained dCGF-IS (per-species coordinates)



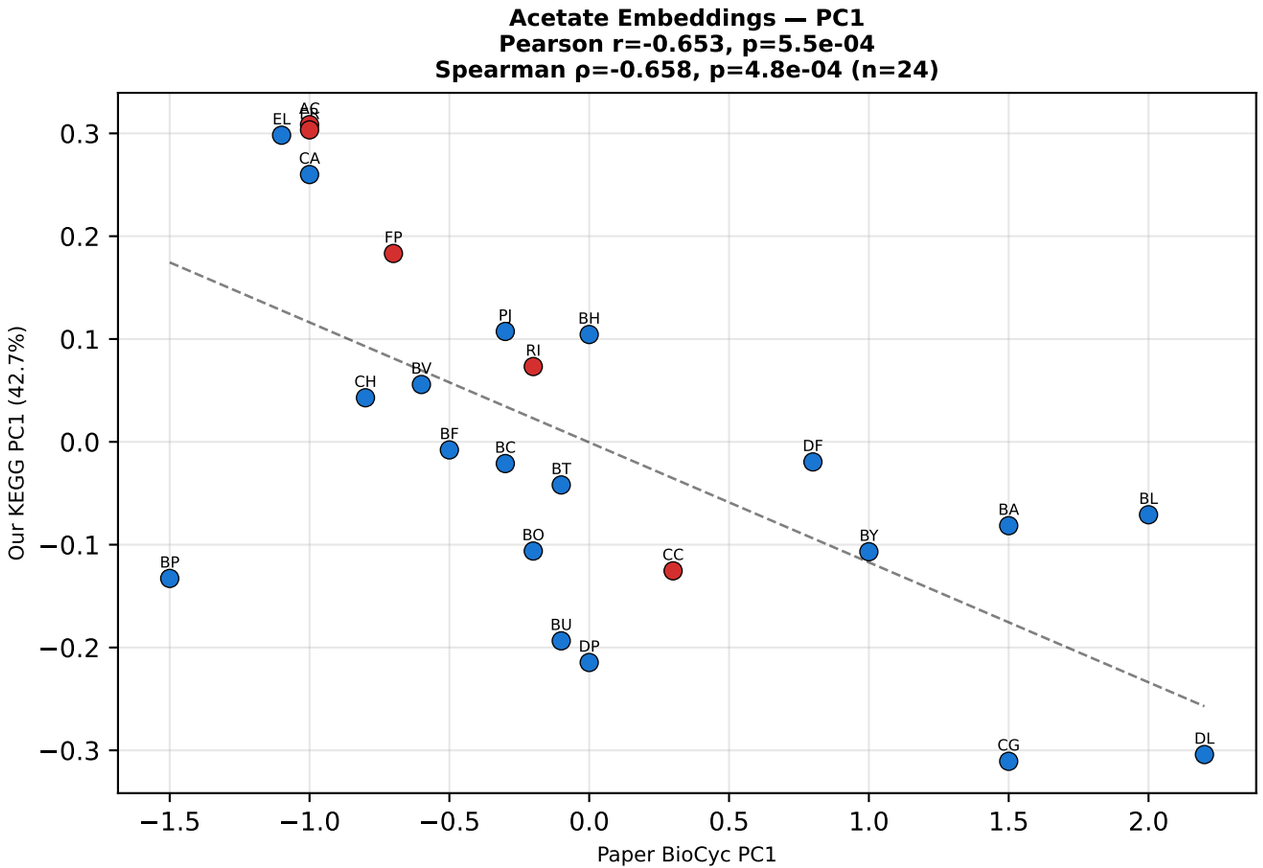


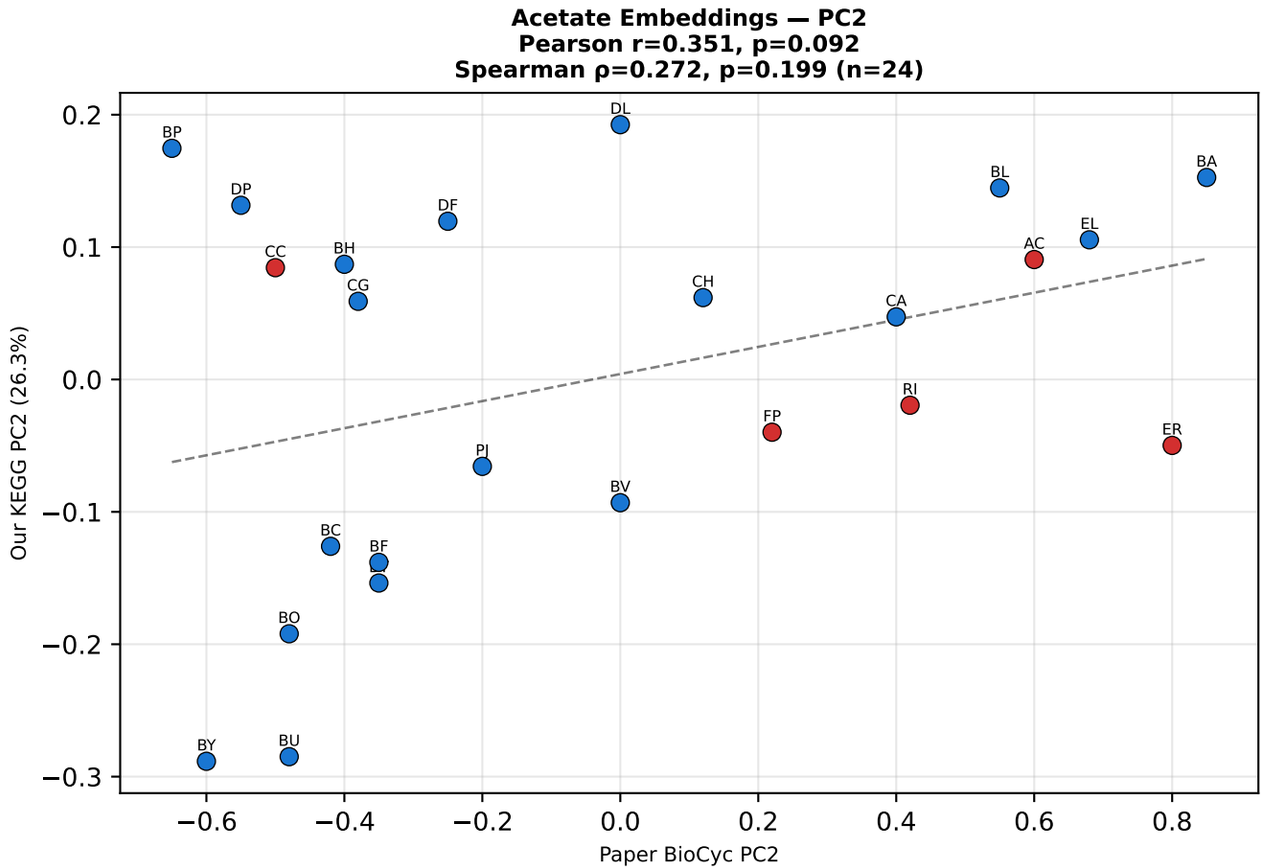
PCA Comparison: Lactate Functional Embeddings
Our KEGG-trained dCGF-IS vs Paper BioCyc-trained dCGF-IS (per-species coordinates)





PCA Comparison: Acetate Functional Embeddings
Our KEGG-trained dCGF-IS vs Paper BioCyc-trained dCGF-IS (per-species coordinates)





Results

1. REPRODUCTION OF PAPER RESULTS

We reproduced the dCGF-IS model using 144 KEGG pathways in place of the paper's 500 BioCyc pathways, applied to the Clark et al. (2021) dataset of 576 communities assembled from 25 bacterial species. Despite using fewer and differently sourced pathways, our model achieves Pearson correlations of r = 0.79-0.92 across the four metabolites (Figures 7-8), comparable to the paper's r = 0.80-0.92. Performance plateaus by 500 epochs and is stable across random seeds (Figure 11).

2. dCGF-ES vs dCGF-IS (Figure 14)

The Enzyme Soup model (ES), which pools all species into a single vector before encoding, achieves r = 0.71-0.90 — consistently lower than IS (r = 0.79-0.92). The largest gaps are for Succinate (+0.071) and Acetate (+0.080), confirming that per-species processing adds value. The paper does not report ES results on real data, only on synthetic data.

3. SHUFFLED-TARGET PERMUTATION TEST (Figure 12)

When metabolite labels are randomly permuted, the model achieves r ~ 0 for all four metabolites (mean r = -0.04 to +0.002, N = 4-10 perms). This confirms that the model captures real biological signal, not artifacts of the training procedure. This provides a complementary validation of the model's biological relevance.

Results (continued)

4. SPECIES-ONLY MODEL (Figures 9-10)

A model using one-hot species identity vectors (24 dimensions, no pathway or genomic information) achieves r = 0.56-0.88 — capturing 72-96% of the full model's performance. For Butyrate (r = 0.88) and Lactate (r = 0.82), species identity alone explains nearly all the variance. Only Acetate shows a substantial drop (r = 0.56 vs 0.79), suggesting pathways are most informative for this metabolite.

5. RANDOM PATHWAYS CONTROL (Figures 9-10)

Replacing real KEGG pathways with random binary vectors (80% shared, 20% random per species) yields r = 0.70-0.89 — between the full model and species-only. This shows that extra input dimensions help the NN, but real pathway biology adds modestly beyond what random features provide. The improvement from real over random pathways is largest for Acetate (0.79 vs 0.70) and Succinate (0.92 vs 0.87).

6. SIMPLE BASELINES (Figure 10)

Random Forest on species presence/absence (24-dim) achieves r = 0.54-0.89, comparable to the dCGF-IS species-only model. Adding pathway sums (RF sp+pw, 168-dim) improves to r = 0.68-0.91. However, adding pairwise pathway similarity (RF sp+sim, 300-dim) does NOT help — it performs equal to or worse than species-only RF. This demonstrates that the value of pathways is their specific content, not the genomic similarity they encode between co-occurring species.

Results (continued)

7. RANDOM ABUNDANCE CONTROL (Figure 10)

The random-abundance control randomizes species abundances while keeping pathway data intact, achieving r = 0.52-0.81. Since Clark et al. used equal initial abundances (1/n), randomization perturbs a uniform distribution, limiting the disruption. Additionally, binary vectors serve as near-unique species fingerprints (100-120 of 144 pathways per species), allowing the model to recover species identity even with random weights.

8. SENSITIVITY ANALYSIS (Figures 19-22) AND SPECIES FINGERPRINT CRITIQUE (Figures 23-26)

The sensitivity analysis method (silence/activate each pathway) tends to highlight pathways present in only 1-3 species. For example, 'Biosynthesis of siderophore group' (present only in AC) ranks among the top hits for 3 of 4 metabolites. Of the top 15 sensitivity hits, 7-10 are unique to a single species.

This pattern arises because silencing a pathway unique to one species effectively alters that species' identity within the community. Widely shared pathways (Glycolysis, TCA cycle, Butanoate metabolism, present in 24/25 species) show lower sensitivity because toggling them in one species has minimal impact on the community total.

Our permutation importance method (Figures 19-22, left panels) offers a complementary view that is less influenced by prevalence, though it still primarily reflects which pathways help the model distinguish species contributions.

Summary and Discussion
Qian et al. (2025) bioRxiv 2025.01.04.631316

1. ADDITIONAL CONTROLS PROVIDE CONTEXT
   The random-abundance control achieves r=0.52-0.82 because binary pathway patterns encode species identity. A shuffled-target permutation test yields r~0, confirming real signal. Together these controls bracket the model's performance and clarify the sources of predictive information.

2. SPECIES COMPOSITION IS THE DOMINANT PREDICTOR
   A model using only species presence/absence (no genomic data) achieves r=0.56-0.88. For Butyrate (r=0.88) and Lactate (r=0.82), species identity captures most of the signal. Pathways provide additional improvement, particularly for Acetate (r=0.56 -> 0.79) and Succinate (r=0.80 -> 0.92).

3. SENSITIVITY ANALYSIS AND PATHWAY PREVALENCE
   The sensitivity analysis method (silence/activate pathways) tends to highlight species-unique pathways. 8/15 top hits are in only 1 species, as silencing a rare pathway alters species identity. Widely shared pathways (Glycolysis, TCA cycle) show lower sensitivity due to redundancy.

4. INTERPRETING THE GENOTYPE-FUNCTION RELATIONSHIP
   The model's predictions are driven largely by community composition. The Deep Set architecture preserves per-species identity, and the homogeneous pathway space (100-120/144 shared) provides modest but real discriminating information beyond species presence/absence alone.

APPENDIX

Reproduction Instructions for Claude Agent

Complete instructions for building all models, databases,
and generating this report from scratch.

A1. Overview & Prerequisites

```
OVERVIEW
========
This appendix provides step-by-step instructions for a Claude
agent (or human) to reproduce all models, control experiments,
interpretability analyses, and this PDF report from scratch.

The project evaluates the dCGF model from Qian et al. (2025,
bioRxiv 2025.01.04.631316) on the Clark et al. (2021) bacterial
community dataset. It trains multiple model variants and controls
to characterize the relative contributions of species
composition and pathway information to model predictions.


PREREQUISITES
=============
1. Python environment with these packages:
   - numpy, scipy, pandas, torch (PyTorch), scikit-learn
   - matplotlib, PyMuPDF (fitz), Pillow (PIL)
   - Standard library: os, json, csv, argparse, collections

2. Input data files (must exist in working directory):
   - MasterDF.csv: Clark et al. 2021 community data
     Columns: Treatment, Rep, OD, Acetate, Butyrate, Lactate,
     Succinate, plus 25 species columns (1/0 presence/absence)
     Species: ER, FP, AC, HB, CC, RI, EL, CH, DP, BH, CA,
     PC, PJ, DL, CG, BF, BO, BT, BU, BV, BC, BY, DF, BL,
     BP, BA

   - genetic_features.csv: Binary species x pathway matrix
     Generated by fetch_kegg_pathways.py (see Step 1)
     Rows: 25 species, Columns: 144 KEGG pathways

   - 2025.01.04.631316v1.full.pdf: The paper PDF (optional,
     for extracting paper figures into the report)

3. SLURM cluster access (partition: Orion) OR local compute
   capable of running PyTorch training (~2-24 hours per step)

4. Internet access for Step 1 only (KEGG REST API)
```

A2. File Inventory

```
PYTHON SOURCE FILES
===================
data_processing.py     Data loading & preparation (shared by all scripts)
  - load_community_data(): reads MasterDF.csv
  - load_genetic_features(): reads genetic_features.csv
  - average_replicates(): groups by species set, averages metabolites
  - prepare_dataset(metabolite): returns GF matrices + standardized targets

dcgf_model.py          PyTorch model definitions
  - dCGF_ES: Enzyme Soup (sum species, then encode)
  - dCGF_IS: Individual Species / Deep Set (encode each, then sum)
  - dCGF_IS_Batched: Optimized IS for variable-size communities

metadata.py            Species lists, community definitions, color schemes

fetch_kegg_pathways.py Fetches KEGG pathways via REST API
  - Output: genetic_features.csv + kegg_pathways_cache.json

train_dcgf.py          Main dCGF-ES and dCGF-IS training
train_dcgf_controls.py Control experiments (actual, random abd, shuffled)
train_species_only.py  Species-only ablation (one-hot, no pathways)
train_random_pathways.py  Random binary pathway control
train_simple_baselines.py Ridge + Random Forest baselines
interpret_dcgf.py      Pathway importance (permutation + sensitivity)
generate_predictions.py  Pre-compute CV predictions -> .npz
generate_report.py     This PDF report generator


SLURM LAUNCHER SCRIPTS
======================
run_dcgf.sh              Main dCGF training (1 job)
run_controls_parallel.sh 48 parallel control jobs
run_baselines.sh         Simple baselines (1 job, ~30 min)
run_interpret.sh         4 parallel interpretation jobs
run_report.sh            PDF report generation (1 job, ~10 min)


GENERATED DATA FILES (outputs)
==============================
genetic_features.csv         Species x pathway binary matrix (Step 1)
kegg_pathways_cache.json     Cached KEGG API responses (Step 1)
cv_predictions.npz           Cross-validated predictions (Step 6)
pathway_importance_*.csv     4 files, one per metabolite (Step 5)
simple_baseline_results.json Ridge + RF results (Step 4)
random_pathway_results.json  Random pathway results (Step 3c)
dcgf_results_report_v1.1.pdf      Final report (Step 7)
```

A3. Execution Steps (1-2)

```
STEP 1: FETCH KEGG PATHWAY DATA
================================
Script:  python -u fetch_kegg_pathways.py
Time:    ~5 minutes (network-bound)
Inputs:  Internet access to https://rest.kegg.jp
Outputs: genetic_features.csv, kegg_pathways_cache.json

What it does:
  - Maps 25 species abbreviations to KEGG organism codes
    (e.g., ER -> ere, FP -> fpr, AC -> aci, BT -> bth, ...)
  - Queries KEGG REST API: /list/pathway/{org_code}
  - Builds binary matrix: 25 species x 144 pathways
  - Caches raw API responses in kegg_pathways_cache.json
  - PC (Prevotella copri) not in KEGG -> row of zeros
  - HB (Holdemanella biformis) may have limited coverage

Note: If genetic_features.csv already exists and you trust it,
skip this step. The cache file allows re-running without API.


STEP 2: TRAIN MAIN dCGF MODELS (ES and IS)
==========================================
Script:  python -u train_dcgf.py
SLURM:   sbatch run_dcgf.sh
Time:    ~2-4 hours
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: Console output with Pearson r for each metabolite

What it does:
  - For each of 4 metabolites (Acetate, Butyrate, Lactate, Succinate):
    * Prepares dataset: loads communities, builds GF matrices,
      averages replicates, standardizes targets
    * Trains dCGF-ES and dCGF-IS with 8-fold CV, 5 random seeds
    * 500 epochs per fold, lr=1e-3, weight_decay=1e-4
    * Reports per-fold and overall Pearson r

Hyperparameters (from paper):
  z_dim=30, encoder_hidden=200, decoder_hidden=100
  k_folds=8, n_seeds=5, n_epochs=500
  lr=1e-3, weight_decay=1e-4

Expected results:
  dCGF-IS: Acetate~0.79, Butyrate~0.92, Lactate~0.87, Succinate~0.92
  dCGF-ES: Acetate~0.71, Butyrate~0.90, Lactate~0.81, Succinate~0.85
```

A4. Execution Steps (3a-3c): Control Experiments

```
STEP 3a: TRAIN CONTROL MODELS (actual, random abd, shuffled)
============================================================
Script:  python -u train_dcgf_controls.py [args]
SLURM:   bash run_controls_parallel.sh  (submits 48 jobs)
Time:    ~8-24 hours (parallelized)
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: Console output per condition

Three conditions, each for 4 metabolites:
  --condition actual:   Baseline replication (should match Step 2)
  --condition random:   Random abundance weights instead of 1/n
  --condition shuffled --perm P:  Shuffle target values (P=0..9)

Evaluates at 500, 1000, and 2000 epochs.
Shuffled condition runs 10 permutations for null distribution.

SLURM parallel submission (48 jobs total):
  4 metabolites x 1 actual = 4 jobs
  4 metabolites x 1 random = 4 jobs
  4 metabolites x 10 shuffled permutations = 40 jobs


STEP 3b: TRAIN SPECIES-ONLY MODEL
=================================
Script:  python -u train_species_only.py
Time:    ~1-2 hours
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: Console output with species-only r per metabolite

Replaces 144-dim pathway vectors with 24-dim one-hot species
identity vectors. Uses dCGF-IS architecture with modified input.
Same hyperparameters: 8-fold CV, 5 seeds, 500 epochs.

Expected: Acetate~0.56, Butyrate~0.88, Lactate~0.82, Succinate~0.80


STEP 3c: TRAIN RANDOM PATHWAYS MODEL
====================================
Script:  python -u train_random_pathways.py
Time:    ~2-4 hours
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: random_pathway_results.json

Generates 144-dim binary vectors: 80% pathways shared by all
species, 20% randomly assigned (binomial p=0.5, seed=12345).
Trains dCGF-IS with these fake pathways.

Expected: Acetate~0.70, Butyrate~0.89, Lactate~0.85, Succinate~0.87
```

A5. Execution Steps (4-5): Baselines & Interpretability

```
STEP 4: TRAIN SIMPLE BASELINES (Ridge + Random Forest)
=======================================================
Script:  python -u train_simple_baselines.py
SLURM:   sbatch run_baselines.sh
Time:    ~10-30 minutes
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: simple_baseline_results.json

Trains 4 models per metabolite (8-fold CV, 5 seeds):
  1. Ridge regression on species presence (24-dim)
     - Alpha=1.0, linear model
  2. Random Forest on species presence (24-dim)
     - 100 trees, default sklearn settings
  3. Random Forest on species + pathway sums (168-dim)
     - Concatenates 24-dim species + 144-dim summed pathways
  4. Random Forest on species + pairwise similarity (300-dim)
     - 24-dim species + 276-dim pairwise Pearson correlations
       of pathway vectors between all species pairs in community

Output JSON structure:
  {metabolite: {ridge: r, random_forest: r,
   rf_species_pathways: r, rf_species_pairwise_sim: r}}


STEP 5: PATHWAY INTERPRETABILITY ANALYSIS
==========================================
Script:  python -u interpret_dcgf.py --metabolite METABOLITE
SLURM:   bash run_interpret.sh  (submits 4 parallel jobs)
Time:    ~4-8 hours per metabolite
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: pathway_importance_{metabolite}.csv (4 files)

Two methods per metabolite:

  Method 1 - Permutation Importance:
    For each of 144 pathways, shuffle that pathway's values
    across species within each community (10 shuffles).
    Importance = base_r - mean(permuted_r).
    Bigger drop = pathway more important for prediction.

  Method 2 - Sensitivity Analysis (paper's Figure 4 method):
    For each pathway in each species: if present, silence it
    (set 1->0); if absent, activate it (set 0->1).
    Sensitivity = |F(perturbed) - F(original)| / delta.
    Average across species, communities, 5-model ensemble.

Output CSV columns: pathway, perm_importance, sensitivity_abs,
                    sensitivity_signed
```

A6. Execution Steps (6-7): Predictions & Report

```
STEP 6: GENERATE CROSS-VALIDATED PREDICTIONS
============================================
Script:  python -u generate_predictions.py
Time:    ~2-4 hours
Inputs:  MasterDF.csv, genetic_features.csv
Outputs: cv_predictions.npz

Re-trains dCGF-IS with 8-fold CV, 5 seeds, 500 epochs.
Collects out-of-fold predictions for every community.
Averages across seeds, unstandardizes to original units.

NPZ keys (per metabolite):
  {metabolite}_measured   - array of measured concentrations
  {metabolite}_predicted  - array of predicted concentrations
  {metabolite}_r          - Pearson correlation coefficient

Note: Succinate uses 2 seeds (speed optimization).


STEP 7: GENERATE PDF REPORT
===========================
Script:  python -u generate_report.py
SLURM:   sbatch run_report.sh
Time:    ~5-10 minutes
Inputs:  cv_predictions.npz
         pathway_importance_*.csv (4 files)
         simple_baseline_results.json
         random_pathway_results.json
         kegg_pathways_cache.json (for pathway prevalence)
         2025.01.04.631316v1.full.pdf (optional, for paper figs)
Outputs: dcgf_results_report_v1.1.pdf

IMPORTANT: All prerequisite files from Steps 1-6 must exist
before running this step. The report reads pre-computed results
and assembles them into figures, tables, and narrative text.


COMPLETE EXECUTION ORDER (SLURM)
================================
  # Step 1: One-time data fetch (run interactively)
  python -u fetch_kegg_pathways.py

  # Steps 2-5: Can run in parallel
  sbatch run_dcgf.sh                  # Step 2: main models
  bash run_controls_parallel.sh        # Step 3a: 48 control jobs
  python -u train_species_only.py      # Step 3b: species-only
  python -u train_random_pathways.py   # Step 3c: random pathways
  sbatch run_baselines.sh              # Step 4: Ridge + RF
  bash run_interpret.sh                # Step 5: interpretability

  # Step 6: After Steps 2-5 complete
  python -u generate_predictions.py

  # Step 7: After Step 6 completes
```

A7. Model Architecture & Hyperparameter Reference

```
dCGF-IS (Individual Species / Deep Set) — PRIMARY MODEL
=========================================================
Parameters: 67,031 total (95% encoder, 5% decoder)

  Input per species i:  [u_i; w] = concat(species_vec, community_sum)
                        u_i = 144-dim pathway vector * (1/n_species)
                        w   = sum of all u_i = 144-dim
                        Concatenated input: 288-dim

  Shared Encoder (applied to EACH species independently):
    Linear(288 -> 200)  = 57,800 params  (288*200 + 200 bias)
    ReLU
    Linear(200 -> 30)   = 6,030 params   (200*30 + 30 bias)
    ReLU
    Encoder total: 63,830 params (95.2%)

  Aggregation: z = sum(z_i for all species in community)

  Decoder:
    Linear(30 -> 100)   = 3,100 params   (30*100 + 100 bias)
    ReLU
    Linear(100 -> 1)    = 101 params      (100*1 + 1 bias)
    Decoder total: 3,201 params (4.8%)


dCGF-ES (Enzyme Soup)
=====================
Parameters: 38,231 total

  Input: w = sum of all species vectors = 144-dim
  Encoder: Linear(144->200)->ReLU->Linear(200->30)->ReLU
  Decoder: Linear(30->100)->ReLU->Linear(100->1)


TRAINING CONFIGURATION (all models)
====================================
  Optimizer: Adam(lr=1e-3, weight_decay=1e-4)
  Loss: MSE on standardized targets
  Epochs: 500 (controls also eval at 1000, 2000)
  CV: 8-fold, 5 random seeds, averaged
  Target standardization: z = (y - mean) / std
  Abundance weighting: equal, a_i = 1/n_species
  Evaluation metric: Pearson r (predicted vs measured)
```

A8. Data Schema & Key Function Reference

```
MasterDF.csv SCHEMA
===================
Columns: [index], Treatment, Rep, OD,
         Acetate, Butyrate, Lactate, Succinate,
         ER, FP, AC, HB, CC, RI, DP, BH, CA, PC, DL, CG,
         BF, EL, CH, BO, BT, BU, BV, BC, BY, PJ, DF, BL,
         BP, BA, [species fraction columns...]
Species columns: 1 = present, 0 = absent
Metabolite columns: concentration in mM
Source: Clark et al. 2021 (Nat Commun)


genetic_features.csv SCHEMA
===========================
Row 1: header = species, pathway_1, pathway_2, ..., pathway_144
Rows 2-26: species_code, 0/1, 0/1, ..., 0/1
144 KEGG metabolic pathways
PC has all zeros (not in KEGG), HB excluded from analysis


KEY DATA PROCESSING FUNCTIONS (data_processing.py)
==================================================
prepare_dataset(data_dir, metabolite, exclude_species):
  1. Loads MasterDF.csv -> list of community dicts
  2. Loads genetic_features.csv -> species pathway vectors
  3. Excludes species without GF data (PC, HB)
  4. Filters communities containing excluded species
  5. Averages replicates (same species set -> average metabolites)
  6. Builds GF matrices: (144, n_species) per community
     Each column = pathway_vector * (1 / n_species)
  7. Standardizes targets: z = (y - mean) / std
  Returns: gf_matrices, targets, mean, std, gf_vectors, pathway_names


SPECIES ABBREVIATIONS
=====================
ER=Eggerthella rectalis    FP=Faecalibacterium prausnitzii
AC=Acidaminococcus          CC=Coprococcus catus
RI=Roseburia intestinalis  EL=Eubacterium limosum
CH=Clostridium hiranonis   DP=Desulfovibrio piger
BH=Blautia hydrogenotrophica  CA=Collinsella aerofaciens
PC=Prevotella copri        PJ=Parabacteroides johnsonii
DL=Dorea longicatena       CG=Clostridium scindens group
BF=Bacteroides fragilis    BO=Bacteroides ovatus
BT=Bacteroides thetaiotaomicron  BU=Bacteroides uniformis
BV=Bacteroides vulgatus    BC=Bacteroides cellulosilyticus
BY=Bifidobacterium longum  DF=Desulfovibrio fairfieldensis
BL=Bacteroides_L           BP=Bacteroides_P
BA=Bacteroides_A           HB=Holdemanella biformis
```

A9. Troubleshooting & Agent Notes

```
COMMON ISSUES
=============
1. KEGG API rate limiting:
   fetch_kegg_pathways.py includes delays between requests.
   If it fails, re-run — it uses cached results for completed species.

2. PC and HB species:
   PC (Prevotella copri) has no KEGG data -> excluded automatically.
   HB (Holdemanella biformis) excluded by default in DCGF_SPECIES.
   Communities containing these species are dropped entirely.
   Effective species count: 23 (of 25 in Clark dataset).

3. Memory/GPU:
   All models run on CPU (no GPU required). PyTorch CPU is sufficient.
   Memory: 8GB is adequate for all scripts.

4. Reproducibility:
   Random seeds are set explicitly in all scripts.
   5 seeds (0-4) are averaged to reduce variance.
   Random pathways use seed=12345 for the binary matrix.

5. CSV parsing:
   pathway_importance_*.csv files may have commas in pathway names.
   Use rsplit(',', 3) to parse from the right (last 3 fields are floats).


NOTES FOR CLAUDE AGENT REPRODUCTION
===================================
- Always use 'python -u' and flush=True for print statements
  (required for SLURM output buffering on Orion partition)

- The project directory should contain MasterDF.csv before starting.
  This file comes from Clark et al. 2021 supplementary data.

- Steps 2-5 are independent and can run in parallel on SLURM.
  Step 6 depends on Step 2 architecture but re-trains from scratch.
  Step 7 depends on outputs from Steps 3c, 4, 5, and 6.

- Total compute time (sequential): ~40-60 hours
  Total compute time (parallel on SLURM): ~8-12 hours

- To add a new control experiment or baseline:
  1. Write a new train_*.py script following existing patterns
  2. Output results as JSON (metabolite -> r dict)
  3. Add reading logic in generate_report.py
  4. Add a new figure/table page to the PDF

- The paper uses 500 BioCyc pathways; we use 144 KEGG pathways.
  Our results are comparable to theirs despite fewer pathways,
  suggesting that 144 KEGG pathways capture the essential
  information needed for metabolite prediction.
```